Adaptive estimation of rare signs with a known finite gap law

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Revision 2: Section 7 added

Abstract

Each fresh coin has an unobserved, persistent Bernoulli bias $1/2+SD$, where the magnitude D has a known finite law and the conditional positive-sign probability is unknown. We characterize the minimax root-mean-square error for estimating the population fraction of positive signs, bounded by a known rarity parameter ρ, under a uniform expected observation budget. Algorithms may create, revisit and interleave coins adaptively. The rate has a closed form: remove the smallest-gap mass ρ, retain its truncated inverse-gap cost, and add the square root of the dyadic inverse-square cost entropy of the remaining mass, scaled by rarity and budget. The lower bound combines a posterior-martingale tail obstruction, entropy packing, a one-coin activation argument and a common-pair reduction valid for fully adaptive population algorithms. The upper bound uses an observable likelihood-ratio classifier race and a finite multilevel Poisson allocation. All specialized ingredients are proved, and explicit universal comparison constants are given. A short deterministic comparison shows that the closed formula is within a factor $1+2\sqrt{2}$ of the entropy–tail selection functional of a companion general-law characterization, so the two rates agree.

1 Statistical model and main theorem

1.1 Persistent coins and the observation budget

Let

$$\mu=\sum_{i=1}^{n} w_i\delta_{d_i}, \qquad 0<d_1<\cdots<d_n\le\tfrac{1}{2}, \qquad w_i>0, \quad \sum_{i=1}^{n} w_i=1,$$

be a known finite probability law. Equal gaps are merged before using this representation. A parameter is a function q on the support of μ. The complete latent pairs (D_k,S_k) of distinct fresh coins are iid, with $D_k\sim\mu$, $S_k\in\{-1,+1\}$, and $\mathbb{P}_q(S_k=+1\mid D_k=d)=q(d)$. Each coin retains its pair on every revisit. Conditional on all latent pairs, the observation tapes of different coins are independent, and tape k consists of iid Bernoulli$(1/2+S_kD_k)$ observations. Neither D_k nor S_k is observed.

Fix a known rarity bound $0<\rho\le 1/4$. The parameter class and target are

$$\mathcal{Q}_\rho=\left\{q:\ 0\le q(d_i)\le 1, \quad \theta(q):=\sum_{i=1}^{n} w_iq(d_i)\le\rho\right\}. \tag{1}$$

This is a global constraint on the positive-sign fraction; it does not impose $q(d_i)\le\rho$ at each gap.

An algorithm may create, revisit and interleave coins, and use an independent random seed whose distribution does not depend on q. Its choices and output are measurable functions of that seed, coin identities and purchased observations. It must terminate and return a real estimate $\widehat{\theta}$ almost

surely under every $q \in \mathcal{Q}_\rho$. Each observed flip costs one, including the first observation of a fresh coin. Let N be the total number of observations. Coin creation, arithmetic and random-number generation are not charged; creating an unobserved coin yields no information. Zero observations are allowed, and no deterministic total cap is imposed.

For a known budget $B > 0$, define the minimax root-mean-square error

$$R_{\mu,\rho}(B) = \inf_{\mathcal{A}:\ \sup_{q\in\mathcal{Q}_\rho} \mathbb{E}_q N \le B} \ \sup_{q\in\mathcal{Q}_\rho} \left\{\mathbb{E}_q(\widehat{\theta} - \theta(q))^2\right\}^{1/2}. \tag{2}$$

The budget is uniform over the entire parameter class. A procedure may depend on (μ, ρ, B), but not on q or any genuine coin's latent state. Projecting its estimate onto $[0, \rho]$ cannot increase loss or work. We abbreviate $R_{\mu,\rho}(B)$ to $R(B)$ when the inputs are fixed.

1.2 The closed quantile functional

Remove exactly the mass ρ with smallest gaps. In terms of the weights, set

$$\tau_i = \min\left\{w_i, \left(\rho - \sum_{\ell<i} w_\ell\right)_+\right\}, \qquad \nu = \sum_i \tau_i \delta_{d_i}, \qquad \kappa = \mu - \nu. \tag{3}$$

Here $x_+ = \max\{x, 0\}$. Let t be the smallest support point for which $\mu((0, t]) \ge \rho$. Then ν has mass ρ and is supported on $(0, t]$, while κ has mass $1 - \rho$ and is supported on $[t, 1/2]$. The atom at t may be split fractionally.

For $j = 0, 1, 2, \ldots$, define the dyadic bands and exact core costs by

$$\begin{gathered} b_j = 2^{-j-1}, \qquad I_j = (b_j/2, b_j], \qquad a_j^* = \sum_{i:d_i\in I_j} \frac{w_i - \tau_i}{d_i^2}, \qquad A^* = \sum_j a_j^*, \\ \Gamma_\kappa = A^* + \sum_{j:a_j^*>0} a_j^* \log(A^*/a_j^*). \end{gathered} \tag{4}$$

All logarithms are natural. There are finitely many positive costs, and $A^* > 0$. Thus $\Gamma_\kappa = A^*(1 + \mathrm{Ent}(a^*/A^*))$ is the inverse-square cost of the core multiplied by one plus the entropy of its distribution across dyadic scales.

With $h = B^{-1/2}$, set

$$T_h = \sum_i \tau_i \min\{1, h/d_i\}, \qquad \mathcal{Q}(B) = \min\left\{\rho,\ T_h + h\sqrt{\rho\Gamma_\kappa}\right\}. \tag{5}$$

The expression depends only on the known weights, gaps, rarity and budget. No optimization over subsets or submeasures is needed to evaluate it.

1.3 The theorem and its quantifiers

Theorem 1.1 (Closed quantile characterization)**.** *For every integer $n \ge 1$, every finite law μ described above, every $0 < \rho \le 1/4$, and every $B > 0$,*

$$10^{-20}\mathcal{Q}(B) \ \le\ R_{\mu,\rho}(B) \ \le\ 30001\mathcal{Q}(B). \tag{6}$$

More precisely, for each (μ, ρ, B) there is one observable algorithm satisfying, simultaneously for all $q \in \mathcal{Q}_\rho$,

$$\mathbb{E}_q N \le B, \qquad \left\{\mathbb{E}_q(\widehat{\theta} - \theta(q))^2\right\}^{1/2} \le 30001\mathcal{Q}(B).$$

Conversely, every algorithm with $\sup_{q\in\mathcal{Q}_\rho} \mathbb{E}_q N \le B$ has worst-case RMS error at least $10^{-20}\mathcal{Q}(B)$, including algorithms with arbitrary adaptive interleaving, nonlinear outputs and unbounded stopping.

The constants are universal and are not optimized. The claim is a comparison up to constants, not an exact formula for the minimax coefficient. The iid assumption concerns the complete latent pairs, not merely their magnitudes. The theorem concerns known finite laws and expected observation budgets; other information structures and budget conventions are not included.

1.4 Notation and testing conventions

Write $s = \mathbf{1}_{\{S=+1\}} \in \{0,1\}$ for a binary sign label; thus the conditional observation mean is $1/2 + (2s-1)d$. A single-unit procedure means a procedure on one genuine coin, with its latent state retained throughout. A null prior is $q \equiv 0$; $\mathbb{E}_0$ denotes expectation at that prior. When the two signs are compared at one specified gap, $\mathbb{E}_{-d}$ and $\mathbb{E}_{+d}$ refer to Bernoulli means $1/2 - d$ and $1/2 + d$, respectively. Fair-coin expectations are explicitly labelled $\mathbb{E}_{\text{fair}}$.

For probability laws P, Q, dominated by a common measure, write

$$\mathrm{Aff}(P,Q) = \int \sqrt{dP\,dQ}, \qquad \mathfrak{h}(P,Q) = 1 - \mathrm{Aff}(P,Q), \qquad \mathrm{TV}(P,Q) = \sup_E |P(E) - Q(E)|.$$

In particular, $\mathfrak{h}$ denotes squared Hellinger distance in the convention used here, whereas $h = B^{-1/2}$ is a budget scale. We use $\mathrm{TV} \le \sqrt{2\mathfrak{h}}$, $\mathfrak{h} \le \mathrm{KL}/2$ and Pinsker's inequality $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$, as well as KL data processing and the finite-prefix chain rule. Unbounded stopping is handled by finite caps and the stated limiting arguments. For a probability vector p, $\mathrm{Ent}(p) = \sum_{p_j>0} p_j \log(1/p_j)$; $\log_+ x = \max\{0, \log x\}$, with $\log_+ 0 = 0$. The symbols $\lesssim$, $\gtrsim$ and $\asymp$ suppress universal positive constants only.

Lemma 1.2 (Two-point RMS testing)**.** *If two admissible parameters have target separation $\Delta > 0$ and terminal transcript laws of total variation at most $1/2$, every estimator has maximum RMS error at those parameters at least $\Delta/4$.*

Proof. Threshold the estimate at the midpoint of the two target values. Each testing error incurs squared estimation loss at least $\Delta^2/4$. The sum of the two testing errors is at least $1 - \mathrm{TV} \ge 1/2$. Hence the sum of the two mean-square errors is at least $\Delta^2/8$, and their maximum is at least $\Delta^2/16$. □

2 Proof strategy and entropy preliminaries

The two terms in (5) describe different obstructions. The removed mass ν may concentrate at gaps too small to resolve at the current budget. Its contribution is the truncated inverse-gap cost T_h. In the remaining mass κ, collective sign uncertainty and identifying the relevant scales produce the cost-entropy term $h\sqrt{\rho\Gamma_\kappa}$.

2.1 Organization of the proof

The lower proof in Section 3 first obtains $R(B) \ge T_h/32$ by a posterior-martingale calculation and an adaptive midpoint decomposition. It then filters the dyadic core costs. Either the tail already controls the desired rate, or the filter preserves a constant fraction of the core entropy. Grouping and trimming the surviving bands produces conditional experiments that satisfy the required rarity and saturation constraints. Their lower bounds add through a common-pair argument.

The upper proof in Section 4 uses the same quantile split as a known design choice. A likelihood bank races an ordinary capped sign classifier on observations of the same coin. Coupled corrections at successive accuracies are replicated with independent Poisson counts. The resulting finite

multilevel estimator has uniform expected work at most B. A homogeneous kernel estimate controls mixed work across scales. Section 5 assembles all branches and tracks the constants in Theorem 1.1. Section 7 compares $\mathcal{Q}$ with the selection functional of a companion general-law manuscript.

Appendix A proves the classification, work and allocation bounds. Appendix B proves weighted sign testing, the activation gate and the annular one-coin obstruction. Appendix C proves finite convex comparison, the common pair for all capped local policies, the adaptive midpoint decomposition and additive localization. No statistical selection theorem is assumed.

2.2 Relation to prior methods

Lee and Valiant [1] study adaptive estimation for mixtures of unknown coins and provide a midpoint decomposition for fully adaptive observations. We reproduce the finite-prefix decomposition needed here, including observable types and the passage to uniform expected budgets. Their sample-complexity theorem is not used as a substitute for the RMS statement above.

The weighted sign-testing argument follows the gap-entropy mechanism of Chen, Li and Qiao [2], including their Bernoulli extension. The specialized separated-gap lemma is proved in Appendix B. The finite convex comparison has methodological antecedents in Juditsky and Nemirovski [3] and Polyanskiy and Wu [4]; its exact version and the simultaneous common-pair construction are proved in Appendix C. We use Hoeffding's inequality [5] and Sion's convex minimax theorem [6], together with the standard probability and compactness facts indicated where applied. The resulting quantile formula and all specialized reductions are established within this manuscript. These attributions identify methodological dependencies and do not assert a comprehensive novelty comparison.

Aronow, Kallus and Lopatto [7] prove the gap-entropy conjecture for fixed-confidence best-arm identification among independent unit-variance Gaussian arms with means in $[0,1]$ and a unique best arm i^*. For algorithms that identify the best arm with probability at least $1-\delta$ on every such instance, in the range of δ treated there, the optimal expected sample count averaged over permutations of the arm labels is within constant factors of $H\,(\log(1/\delta)+\mathrm{Ent}(\pi))$. Here $H=\sum_{i\neq i^*}\gamma_i^{-2}$, γ_i is the gap from arm i to the best mean, and π gives the shares of H in dyadic gap ranges. The core term Γ_κ has the same form. That problem identifies one of finitely many labelled arms with a confidence requirement; here the target is a population fraction under RMS loss and an expected budget, with fresh unlabelled coins and unknown signs. We do not use their result and do not claim that either result implies the other.

The companion manuscript [8] states a characterization of $R_{\mu,\rho}(B)$ for every law on $(0,1/2]$ by a functional $\mathcal{K}$ that optimizes over submeasures of μ; its proof has not been externally refereed or formally verified. Section 7 shows that $\mathcal{Q}$ and $\mathcal{K}$ agree within the factor $1+2\sqrt{2}$. Theorem 1.1 is therefore the finite-law case of that stated characterization, here with a closed functional and a self-contained proof that does not invoke the theorem of [8]. Both manuscripts were developed from the same research notes: the two proofs share the observable upper construction and most appendix lemmas and differ mainly in the lower-bound reduction. The comparison itself uses neither statistical theorem.

2.3 Cost entropy and canonical band costs

For a finite nonnegative vector a, put $A=\sum_j a_j$ and define $\mathcal{G}(a)=A+\sum_{j:a_j>0}a_j\log(A/a_j)$, with $\mathcal{G}(0)=0$. This continuous function is homogeneous, coordinatewise increasing and concave. On the positive coordinates, its derivative is $1+\log(A/a_j)$ and its Hessian is $A^{-1}\mathbf{1}\mathbf{1}^\top-\mathrm{diag}(1/a_j)$. Cauchy–Schwarz makes the latter negative semidefinite, and continuity extends the inequalities

to zero coordinates. Homogeneity and concavity imply superadditivity, and binary entropy gives

$$\mathcal{G}(a+b) \le (1+\log 2)[\mathcal{G}(a)+\mathcal{G}(b)] \le 2[\mathcal{G}(a)+\mathcal{G}(b)]. \tag{7}$$

Indeed, keep and then forget the label specifying which cost vector generated a coordinate. The extra entropy is at most $\log 2$ times the total cost, which is at most $\mathcal{G}(a)+\mathcal{G}(b)$.

In the lower bound we use canonical costs

$$v_j = \kappa(I_j), \quad a_j = v_j/b_j^2, \quad A = \sum_j a_j, \quad F = \mathcal{G}(a), \quad \lambda = \sqrt{\rho/F}.$$

The core has mass $1-\rho > 0$, so $F > 0$. Coordinatewise comparison yields

$$F \le \Gamma_\kappa \le 4F. \tag{8}$$

Only costs are rounded; the observation experiment retains its actual gaps.

3 Lower bound

Throughout the proof, $h = B^{-1/2}$ is the budget scale and $\mathfrak{h}(P,Q)$ is the squared Hellinger distance defined in Section 1.4.

3.1 An explicit posterior-martingale bound for the quantile tail

Fix a gap d and a capped one-coin policy. Under the equal sign prior, let $U_k = \mathbb{E}[S \mid \text{seed}, X_1, \dots, X_k]$, with $X_i \in \{-1, 1\}$ and $\mathbb{E}[X_i \mid S] = 2dS$. Then $U_0 = 0$ and U_k is a martingale. For a current posterior mean u,

$$U_{k+1} = \frac{u + 2dX_{k+1}}{1 + 2duX_{k+1}}, \qquad \mathbb{E}[(U_{k+1}-u)^2 \mid U_k = u] = \frac{4d^2(1-u^2)^2}{1-4d^2u^2} \le 4d^2.$$

At $d = 1/2$, the posterior becomes absorbing at ± 1 after the first observation; the undefined zero-over-zero expression there is interpreted through the zero increment. For the policy's capped stopping time N,

$$\mathbb{E}_{1/2} U_N^2 \le 4d^2 \mathbb{E}_{1/2} N. \tag{9}$$

The seed may be arbitrary and independent of the latent state.

Change the conditional positive-sign probability to $1/2-\epsilon$ or $1/2+\epsilon$, $0 \le \epsilon \le 1/4$. Relative to the midpoint transcript law the endpoint densities are $1-2\epsilon U_N$ and $1+2\epsilon U_N$. Since $1-\sqrt{1-x} \le x$ on $[0,1]$, (9) gives

$$\mathfrak{h}(P_{1/2-\epsilon}, P_{1/2+\epsilon}) \le 4\epsilon^2 \mathbb{E}_{1/2} U_N^2 \le 16\epsilon^2 d^2 \mathbb{E}_{1/2} N. \tag{10}$$

Lemma 3.1 (Quantile-tail obstruction)**.** *For every allowed law, rarity and budget, $R_{\mu,\rho}(B) \ge T_h/32$.*

Proof. Use an oracle experiment that lets the algorithm choose any tail atom i as an observable type, then obtain a genuine coin with that fixed gap d_i and unknown sign. Each flip is still charged. The algorithm may interleave types and revisit coins. Set

$$\epsilon_i = \frac{1}{16}\min\{1, h/d_i\}, \qquad q_{i,\pm} = 1/2 \pm \epsilon_i.$$

Equation (10) bounds the local distance for every type by $(h^2/16)$ times its midpoint work. The midpoint decomposition, proved in Lemma C.2, therefore bounds global distance by $h^2 B/16 = 1/16$. It permits observable types chosen adaptively and arbitrary interleaving.

Every original algorithm can be simulated in this oracle experiment: for each fresh coin privately choose tail type i with probability τ_i; on the remaining mass simulate an all-negative coin with the residual magnitude law $\kappa/(1-\rho)$ for free. Keep that private choice on revisits. The resulting original endpoints are

$$q_{\pm}(d) = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}(d)\left(\frac{1}{2} \pm \frac{1}{16}\min\{1, h/d\}\right).$$

The density uses total masses at a support point. At a distinct atom it equals τ_i/w_i; this remains well-defined even if an input lists identical gaps more than once. Both endpoints and their midpoint are admissible, since their total positive mass is at most $(9/16)\rho$. The oracle's genuine work is at most the original flip count, so its midpoint budget is at most B. No budget assumption over arbitrary oracle parameters is needed. The original target separation is $2\sum_i \tau_i\epsilon_i = T_h/8$. Data processing gives total variation at most $\sqrt{2/16} < 1/2$. Two-point midpoint testing gives RMS at least one quarter of the separation, namely $T_h/32$. For unbounded original stopping, cap the global experiment first. The same endpoint pair works at every cap; increasing prefixes give the terminal TV bound, as proved after Lemma C.2. □

Lemma 3.2 (Rare Bernoulli-label floor)**.** $R_{\mu,\rho}(B) \geq \min\{\rho, h\sqrt{\rho}\}/32$.

Proof. Restrict to constant priors $q(d) = p$ and strengthen the experiment by revealing a fresh coin's sign for unit cost. Its magnitude and subsequent flips can then be simulated using independent randomness, so the number of purchased labels is at most the original number of flips. Take $p_- = \rho/2$, $p_+ = p_- + \delta$ with $\delta = \min\{\rho, h\sqrt{\rho}\}/8$. One Bernoulli-label divergence is at most $\delta^2/[p_+(1-p_+)] \leq 3\delta^2/\rho$. The stopped chain rule bounds total KL by $3\delta^2 B/\rho \leq 3/64$. Pinsker gives TV below $1/2$, and midpoint testing gives RMS at least $\delta/4$. Finite prefixes justify unbounded stopping. □

3.2 A deterministic entropy filter

Retain the canonical core notation a, A, F, λ from Section 2.3. If $h \geq t$, then $T_h = \rho$, so the tail lemma already proves the desired lower bound. Assume $h < t$. If $\lambda \geq t/4$, then

$$h\sqrt{\rho\Gamma_\kappa} \leq 2h\sqrt{\rho F} = 2h\rho/\lambda \leq 8h\rho/t \leq 8T_h,$$

and $\mathcal{Q} \leq 9T_h$. Thus it remains to treat $\lambda < t/4$.

Set $L_j = 1 + \log(A/a_j)$ on positive-cost bands. Declare a band *good* exactly when

$$b_j \geq \tfrac{\lambda}{2} L_j. \tag{11}$$

This is a direct test of the known weights and endpoints. Let F_g and F_b be the entropies of the good and bad cost vectors, each normalized by its own total cost.

Lemma 3.3 (Filtering retains core entropy)**.** *If $\lambda < t/4$, then $F_b \leq F/8$ and $F_g \geq F/4$.*

Proof. Every core band has endpoint at least t. Number the bands upward from the one containing t, with endpoints $\beta_0 2^k$, $k \geq 0$. On a bad band, $L_j > x_j := 2b_j/\lambda$, and $x_j \geq 8$. Since $a_j/A = e^{1-L_j}$ and xe^{1-x} decreases for $x \geq 1$,

$$\sum_{j \text{ bad}} a_j L_j \leq A \sum_{k\geq 0} x_0 2^k e^{1-x_0 2^k}, \qquad x_0 = 2\beta_0/\lambda \geq 8.$$

The ratio of successive summands is at most $2e^{-8} < 1/2$. Their sum is at most $2x_0 e^{1-x_0} \leq 16e^{-7} < 1/8$. The bad vector's own entropy is at most this full-normalization sum, so $F_b \leq A/8 \leq F/8$. By (7), $F \leq 2(F_g + F_b)$; consequently $F_g \geq 3F/8 \geq F/4$. □

3.3 Packing the filtered entropy into actual experiments

We first group coordinates with comparable costs.

Lemma 3.4 (Entropy tiers)**.** *For a finite positive vector a, let $M = \max_j a_j$ and group its coordinates by $a_j \in (M2^{-k-1}, M2^{-k}]$, $k \geq 0$. If F_k is each group's entropy cost, then*

$$\mathcal{G}(a) \leq 7 \sum_k F_k. \tag{12}$$

Proof. Let A_k be the cost mass in group k, n_k its cardinality, and $p_k = A_k/A$. Entropy decomposition gives $\mathcal{G}(a) = \sum_k F_k + A \operatorname{Ent}(p)$. Empty groups are omitted. For the comparison on $k \geq 0$, the reference probability is $q_k = 2^{-k/2}/(2+\sqrt{2})$, so $\mathrm{KL}(p\|q) \geq 0$ applies even when the nonempty tier indices are sparse. Comparison with the probability vector proportional to $2^{-k/2}$ gives

$$A \operatorname{Ent}(p) \leq A \log(2+\sqrt{2}) + \frac{\log 2}{2} \sum_k k A_k.$$

On groups with $n_k \geq 2^{k/2}$, the second term is at most $\sum_k A_k \log n_k \leq \sum_k F_k$; comparable weights imply $F_k \geq A_k[1 + \log n_k - \log 2] \geq A_k \log n_k$. On the other groups, $A_k \leq M2^{-k/2}$, so their contribution is at most

$$\frac{\log 2}{2} M \sum_{k \geq 0} k 2^{-k/2} = \frac{(\log 2)\sqrt{2}}{2(\sqrt{2}-1)^2} M < 3M \leq 3A.$$

Finally $\log(2+\sqrt{2}) < 2$ and $A \leq \sum_k F_k$. Adding the initial $\sum_k F_k$ proves (12). □

Split the good bands into high bands $j < 6$ and low bands $j \geq 6$, with entropies F_H, F_L. Since $b_j \geq 1/64$ on the first six bands, $F_H \leq 4096(1 + \log 6) =: C_H$. If $F_H \geq F_L$, then

$$F \leq 4F_g \leq 8(F_H + F_L) \leq 16 C_H,$$

and the Bernoulli-label floor implies $R \gtrsim \min\{\rho, h\sqrt{\rho F}\}$.

Otherwise $F_L > F_H$, so $F \leq 16 F_L$. Split the low bands by index parity, then within each parity into the comparable-cost tiers of Lemma 3.4. The sum of the resulting group entropies is at least $F_L/14$. Indeed $F_L \leq 2(F_{\text{even}} + F_{\text{odd}})$, and the tier lemma is applied separately within each parity. Both parities and all nonempty tiers are retained; choosing just one tier would not justify this bound. Each group has endpoint ratios at least four and costs in $(c, 2c]$ for some $c > 0$.

For a group of n bands, (11) and $A \geq nc$ give

$$b_j \geq \frac{\lambda}{2}[1 + \log(n/2)] \geq \frac{\lambda}{8}(1 + \log n).$$

Remove its $s = \lceil \log_4 n \rceil$ smallest-gap bands, with $s = 0$ for $n = 1$. For $n \geq 2$, $s \leq n/2$, so at least $n/2$ remain. Let G_i be the remaining group's entropy. Coordinate monotonicity yields

$$G_i \geq F_{\text{group}}/16, \qquad b_{\min,i} \geq \lambda n(1 + \log n)/8. \tag{13}$$

For the first bound, divide the lower bound $cn'(1 + \log n')$ by the upper bound $2cn(1 + \log n)$; the ratio is at least $(1 - \log 2)/4 > 1/16$. The second bound follows because deleting s smallest bands raises the minimum endpoint by at least $4^s \geq n$. The case $n = 1$ is immediate. The integer inequality for $n \geq 2$ can be checked directly at $n = 2, 3, 4$; when $4^{s-1} < n \leq 4^s$ and $s \geq 2$, we have $2s \leq 4^{s-1} + 1 \leq n$. Trimming is by increasing gap (equivalently, increasing endpoint), not by cost. The number n in (13) is the pre-trim cardinality.

Let σ_i be the restriction of the actual core measure to a remaining group, m_i its mass, u_i its largest endpoint, and $w_{*,i}$ its smallest band mass. Then

$$m_i/u_i^2 \geq c, \qquad w_{*,i} \geq c b_{\min,i}^2, \qquad G_i \leq 2cn(1+\log n).$$

Consequently

$$\frac{m_i w_{*,i}}{u_i^2} \geq \lambda^2 G_i^2/256. \tag{14}$$

Assign original rarity $\theta_i = \lambda^2 G_i/1024$ and conditional rarity $r_i = \theta_i/m_i$. Equation (14) gives $\theta_i G_i \leq m_i w_{*,i}/(4u_i^2)$. Because $G_i \geq m_i/u_i^2$, this actually implies $\theta_i \leq w_{*,i}/4 \leq m_i/4$, hence $r_i \leq 1/4$. Here $w_{*,i}$ is a band mass, not a minimum individual atom mass. Under the conditional law σ_i/m_i, the entropy and minimum band mass are G_i/m_i and $w_{*,i}/m_i$. Its entropy-branch condition is

$$r_i u_i^2 (G_i/m_i) \leq w_{*,i}/m_i \quad \Longleftrightarrow \quad \theta_i G_i \leq m_i w_{*,i}/u_i^2.$$

Thus the preceding factor-four margin verifies the actual conditional condition. There is no unnormalized-to-conditional mass factor missing. Disjointness and superadditivity give

$$\sum_i \theta_i \leq \lambda^2 F/1024 = \rho/1024, \qquad \sum_i G_i \geq F_L/224 \geq F/3584.$$

All remaining bands have largest endpoint at most $1/128$ and endpoint ratios at least four. Apply the annular one-unit obstruction of Proposition B.4 to each normalized component σ_i/m_i, followed by the additive localization theorem (Theorem C.4). Their complete proofs are included in Appendices B and C. The actual minimum-band-mass condition is verified by (14), so the entropy branch is active. We obtain

$$R(B) \gtrsim \sum_i \min\{\theta_i, h\sqrt{\theta_i G_i}\} = \left(\sum_i G_i\right) \min\{\lambda^2/1024, h\lambda/32\} \gtrsim \min\{\rho, h\sqrt{\rho F}\}.$$

The proportional allocation $\theta_i/G_i = \lambda^2/1024$ is what makes this estimate valid jointly at every budget.

3.4 Completion of the lower bound

The cases $h \geq t$ and $\lambda \geq t/4$ were handled by the tail lemma. In the remaining case, the floor or the packing argument gives $R \gtrsim \min\{\rho, h\sqrt{\rho F}\}$, while always $R \geq T_h/32$. Finally, by (8),

$$\mathcal{Q}(B) \leq T_h + 2\min\{\rho, h\sqrt{\rho F}\}.$$

The two lower bounds control this sum up to a universal constant. Thus $R(B) \geq c\mathcal{Q}(B)$, proving the lower half of Theorem 1.1. Section 5 tracks the constants to obtain the stated value 10^{-20}.

4 Observable upper bound

We construct an estimator at the quantile split from Section 1. The classification and allocation lemmas used below are proved in Appendix A.

Theorem 4.1 (Upper bound at the fixed quantile split)**.** *For every admissible (μ, ρ, B), there is an observable estimator with uniform expected work at most B and worst RMS error at most $30001\mathcal{Q}(B)$.*

If $\mathcal{Q}(B) = \rho$, use the constant estimator $\rho/2$. Otherwise $h < t$, because $h \geq t$ would imply $T_h = \rho$. Fix

$$\sigma = \kappa, \qquad \tau = \nu|_{[h,1/2]}, \qquad \zeta = \nu|_{(0,h)}, \qquad z = \zeta((0,1/2]), \quad H = \int d^{-1}\,\tau(dd).$$

Then $\mu = \sigma + \tau + \zeta$ and $T_h = z + hH$. These are analytical pieces only. The estimator never observes which piece produced its genuine coin. All likelihood evaluations and design parameters use the known law. The complexity measure counts physical coin observations, including those given to either side of the race; it does not assert a bound on arithmetic time, memory, or random-number generation.

4.1 An uncapped reference bank with a capped observable race

Use the fixed decomposition above, put $z = \zeta((0,1/2])$, and assume $\Gamma_\sigma, H_\tau < \infty$. It suffices to treat $z \leq \rho/128$: otherwise the constant $\rho/2$ has risk at most $64z$. For a finite submeasure ξ, define explicitly

$$a_j(\xi) = \int_{I_j} d^{-2}\,\xi(dd), \qquad A_\xi = \sum_j a_j(\xi), \qquad H_\xi = \int d^{-1}\,\xi(dd),$$

$$\Gamma_\xi = A_\xi + \sum_j a_j(\xi)\log(A_\xi/a_j(\xi)).$$

Zero coordinates are omitted, and all three quantities are zero for the zero measure. These are exact inverse-square costs, not the rounded canonical costs used in the lower bound. Write $A = A_\sigma$, $H = H_\tau$. If $\sigma \neq 0$, use all its occupied bands, with weights $\pi_j = a_j(\sigma)/A$ and references $b_j/2$ and $b_j/4$. For an internal allowance $u < \rho$, set $\epsilon = u/\rho$. Stop negative if a negative reference likelihood against fair reaches $1/(\epsilon\pi_j)$; stop positive if

$$\frac{\sum_j \pi_j P_{1/2+b_j/4,n}}{\int P_{1/2-d,n}\,\mu(dd)} \geq 1/u. \tag{15}$$

Give negative crossings priority. A zero denominator triggers a positive stop. The known all-negative mixture in the denominator is the full law μ, regardless of the decomposition.

By Lemma A.3, a wrong negative crossing has conditional positive-sign probability at most ϵ, and a positive crossing has all-negative-mixture probability at most u. Thus the total probability of a wrong bank stop is at most $2u$ over $\mathcal{Q}_\rho$. Termination outside σ is not assumed.

Race this bank against the ordinary classifier of Lemma A.1, using alternating observations of the same physical coin. At an ordinary cap M, its conditional work certificate is

$$o_M(d) = \min\{M, 17d^{-2}[1 + \log_+(Md^2)]\}.$$

For any finite measure ξ, put

$$\eta_\xi(M) = \int \min\{1, (Md^2)^{-2}\}\,\xi(dd), \qquad O_\xi(M) = \int o_M(d)\,\xi(dd).$$

The ordinary error under μ is at most $\eta_\mu(M)$. Stop as soon as either component stops and return that output. On completed counterfactual streams, the physical work is at most

$$2\min\{T_{\text{bank}}, T_{\text{ordinary}}\} \leq 2M.$$

A wrong output implies either a wrong bank stop or an ordinary error. When $\sigma = 0$, omit the bank and use the ordinary classifier alone.

On a σ-almost-every gap d in band j, the matching negative reference has drift at least $4d^2/3$ and log overshoot below one. Its expected crossing time is at most

$$\frac{3}{4d^2}[1 + \log(1/\epsilon) + \log(1/\pi_j)]. \tag{16}$$

On a positive gap, the event $\widehat{p}_n - 1/2 \geq d/2$ gives

$$\frac{P_{-,\mu,n}}{P_{\mathrm{f},n}} \leq \int e^{-2nx^2}\,\mu(dx) \leq z + \frac{A}{2en} + \frac{H}{2\sqrt{en}}. \tag{17}$$

The last two bounds use, respectively, $\sup_x x^2 e^{-2nx^2} = 1/(2en)$ and $\sup_x x e^{-2nx^2} = 1/(2\sqrt{en})$. For $n \geq d^{-2}$ the denominator is at most $z + Ad^2 + Hd$, while the matching positive reference contributes at least $\pi_j e^{nd^2/3}$ relative to fair. Equation (35) therefore bounds positive expected bank work by

$$\frac{6}{d^2}\left[1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+ \frac{z + Ad^2 + Hd}{\rho}\right]. \tag{18}$$

These arguments prove finite conditional bank work on σ. On $\tau + \zeta$, the ordinary cap guarantees termination of the race.

Since $z/\rho \leq 1$,

$$\log_+ \frac{z + Ad^2 + Hd}{\rho} \leq \log 3 + \log_+(Ad^2/\rho) + \log_+(Hd/\rho).$$

Using $\sup_{x\geq 1} \log x/x = 1/e$ and $\sup_{x\geq 1} \log x/x^2 = 1/(2e)$ gives

$$\int q(d) d^{-2} \log_+ \frac{z + Ad^2 + Hd}{\rho}\,\sigma(dd) \leq (\log 3 + 1/e)A + \frac{H^2}{2e\rho}. \tag{19}$$

Here $\int q\,d\sigma \leq \rho$ and $q \leq 1$ follow from the original global rarity class; no separate conditional rarity assumption is used. Integrating (16) and (18) therefore bounds the bank work on σ by

$$6\Gamma_\sigma + 6A\log(1/\epsilon) + 6(\log 3 + 1/e)A + \frac{3H^2}{e\rho}. \tag{20}$$

4.2 A resolved-law cap schedule

Let $\chi = \sigma + \tau$ and set

$$\delta_\ell = \rho 16^{-(\ell+1)}, \qquad u_\ell = \delta_\ell/4, \qquad \epsilon_\ell = u_\ell/\rho.$$

Choose M_ℓ as the least power of four, at least four, such that $\eta_\chi(M_\ell) \leq \delta_\ell/8$. Every chosen cap has a predecessor because $\eta_\chi(4) = 1 - z$. The inequality $\eta_\chi(4M) \geq \eta_\chi(M)/16$ implies

$$\delta_\ell/128 < \eta_\chi(M_\ell) \leq \delta_\ell/8. \tag{21}$$

Thus the selected powers of four are distinct and increasing.

If $z > 0$, let J be the largest integer with $\delta_J \geq 8z$. Then

$$8z \leq \delta_J < 128z. \tag{22}$$

For $\ell \leq J$, $\eta_\mu(M_\ell) \leq \eta_\chi(M_\ell) + z \leq \delta_\ell/4 = u_\ell$. The raced classifier has total error at most $3u_\ell \leq \delta_\ell$ and false-positive probability at most $2u_\ell \leq \delta_\ell$. When $z = 0$, these conclusions hold at every level.

Since $\log(1/\epsilon_\ell) = \log 64 + \ell \log 16$ and $A \le \Gamma_\sigma$, (20) is at most $40(\ell+1)\Gamma_\sigma + 2H^2/\rho$. A uniform raced-work certificate is consequently

$$C_\ell = 80(\ell+1)\Gamma_\sigma + 4H^2/\rho + 2O_\tau(M_\ell) + 2zM_\ell. \tag{23}$$

It remains a valid, possibly loose, bound when $\sigma = 0$ and the bank is omitted.

The binary level-zero output has second moment at most $\rho + \delta_0 = 17\delta_0$. An adjacent difference, run on successive unused blocks of the same coin, has second moment at most $\delta_\ell + \delta_{\ell-1} = 17\delta_\ell$. Set $v_\ell = 17\delta_\ell$, $k_0 = C_0$, $k_\ell = C_\ell + C_{\ell-1}$ for $\ell \ge 1$, and $S_J = \sum_{\ell \le J} \sqrt{v_\ell k_\ell}$. Independent-Poisson allocation across these corrections (Lemma A.5) gives RMS at most

$$\sqrt{\delta_J^2 + S_J^2/B}. \tag{24}$$

Every such finite-level procedure has finite deterministic local caps. In the finite-law theorem the reference bank is finite; its stopping events are measurable and its positive reference is a finite probability mixture. No countable-support extension is needed here.

4.3 Controlling mixed work without an extra logarithm

For every finite τ, the following bounds hold:

$$O_\tau(M) \le 21H\sqrt{M}, \qquad \eta_\sigma(M) \le A/M, \qquad \eta_\tau(M) \le H/\sqrt{M}. \tag{25}$$

The latter two are pointwise inequalities before integration. For the first, put $x = \sqrt{M}\, d$. The ratio of $o_M(d)$ to $\sqrt{M}/d$ is at most one for $x \le 1$, and at most $17(1 + 2\log x)/x \le 34/\sqrt{e} < 21$ for $x \ge 1$. Lemma A.2 supplies the homogeneous ordinary kernel bound:

$$\sum_{k \ge 1} \sqrt{\eta_\tau(4^k) O_\tau(4^k)} \le 208H. \tag{26}$$

Its proof applies without normalization of τ: dyadic binning yields the summable kernels 2^{k-j} for $j \ge k$, and $2^{3(j-k)}$, $2^{j-k}(1+k-j)$ for $j < k$. Apply $\sqrt{xy} \le (x+y)/2$ and sum these kernels.

Split levels according to which term of $\eta_\chi = \eta_\sigma + \eta_\tau$ is at least half the total. On levels where η_τ dominates, $\delta_\ell \le 256\eta_\tau(M_\ell)$, so distinctness of the caps and (26) give a total $\sum \sqrt{\delta_\ell O_\tau(M_\ell)} \le 3328H$. On the other levels,

$$\delta_\ell \le 256\eta_\sigma(M_\ell) \le 256A/M_\ell, \qquad M_\ell \le 256A/\delta_\ell.$$

Therefore

$$\sqrt{\delta_\ell O_\tau(M_\ell)} \le 4\sqrt{21}\, H^{1/2} A^{1/4} \delta_\ell^{1/4}.$$

The sum of $\delta_\ell^{1/4}$ over all levels is $\rho^{1/4}$. The arithmetic–geometric mean inequality proves

$$\sum_{\ell \le J} \sqrt{\delta_\ell O_\tau(M_\ell)} \le C_{\mathrm{mix}}[H + \sqrt{\rho A}]. \tag{27}$$

The work on the unresolved mass is controlled separately. On levels where η_σ dominates,

$$\sum \sqrt{z\delta_\ell M_\ell} \le 16(J+1)\sqrt{zA} \le \sqrt{2\rho A}. \tag{28}$$

The last inequality uses $z \le \delta_J/8$ and $(J+1)4^{-(J+1)} \le 1/4$. On levels where η_τ dominates, $\sqrt{M_\ell} \le 256H/\delta_\ell$, so

$$\sum \sqrt{z\delta_\ell M_\ell} \le 256H\sqrt{z} \sum_{\ell=0}^{J} \delta_\ell^{-1/2} \le \frac{1024}{3} H\sqrt{z/\delta_J} \le \frac{1024}{3\sqrt{8}} H. \tag{29}$$

When $z = 0$ these contributions vanish.

Completion of Theorem 4.1. The shifted adjacent-level sum has a factor $1/4$, so

$$S_J \le \frac{5\sqrt{17}}{4} \sum_{\ell=0}^{J} \sqrt{\delta_\ell C_\ell}.$$

Use (23), (27), (28), (29), and $A \le \Gamma_\sigma$. The elementary sums $\sum_{\ell\ge 0} 4^{-\ell}\sqrt{\ell+1} \le 16/9$ and $\sum_{\ell\ge 0} 4^{-\ell} = 4/3$ then give

$$S_J \le C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H].$$

One valid explicit choice is $C_{\mathrm{sum}} = 30000$. Indeed, writing $R = \sqrt{\rho\Gamma_\sigma}$, the component bounds above give

$$\sum_{\ell\le J} \sqrt{\delta_\ell C_\ell} \le 21R + \frac{15535}{3} H.$$

Here the core term is at most $4R$, the fixed tail surcharge is $2H/3$, the ordinary mixed term is at most $15R + 5007H$, and the unresolved term is at most $2R + 512H/3$. Use $\sqrt{21} < 5$, $\sqrt{2} < 3/2$ and $5\sqrt{17}/4 < 165/32$ to obtain

$$S_J \le \frac{3465}{32} R + \frac{854425}{32} H \le 30000(R + H).$$

For $z > 0$, (22) and (24) imply $R(B) \le 128z + C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H]/\sqrt{B}$. For $z = 0$, put $D = \sqrt{\rho\Gamma_\sigma} + H > 0$ and choose the least finite $J \ge 0$ such that $\delta_J \le D/\sqrt{B}$. The bound on S_J holds at this finite terminal level. Equation (24) therefore gives RMS at most

$$\sqrt{1 + 30000^2}\, D/\sqrt{B} < 30001 D/\sqrt{B}.$$

Thus this branch has an actual finite-level estimator; no limiting or infinite-level construction is needed. The constant estimator covers $z > \rho/128$. For the fixed split, $z + hH = T_h$ and $\sigma = \kappa$. The preceding bound and the constant estimator therefore give $R(B) \le C \min\{\rho, T_h + h\sqrt{\rho\Gamma_\kappa}\} = C\mathcal{Q}(B)$. The preceding constants permit $C = 30001$ for this upper bound; no optimality of that constant is claimed. The number of levels and every local cap are finite, while the total Poisson replication count is unbounded but almost surely finite and has the stated expected budget. Since the law is finite, the reference bank is finite throughout. □

5 Completion of the main theorem

The same finite known law μ, rarity bound ρ, parameter class $\mathcal{Q}_\rho$, persistent latent-pair experiment, target $\theta(q)$, RMS loss, and uniform expected observation budget are used on both sides. For every fixed (μ, ρ, B), the upper proof gives one observable algorithm whose work and risk bounds hold for every $q \in \mathcal{Q}_\rho$. Its design may depend on (μ, ρ, B) but not on q or a genuine coin's latent state. The lower proof applies to every such algorithm, including nonlinear outputs and arbitrary interleaving. There is no claim of a single procedure simultaneously attaining every budget, or of a deterministic total observation cap.

5.1 Upper-bound branches and actual attainment

Put $D = \sqrt{\rho\Gamma_\kappa} + H$ for the split in Section 4. The following exhaustive branches are selected using only known design parameters:

1. If $\mathcal{Q} = \rho$, output $\rho/2$ at zero cost.
2. If $\mathcal{Q} < \rho$ and $z > \rho/128$, the same constant has risk $\rho/2 < 64z \le 64\mathcal{Q}$.

3. If $\mathcal{Q} < \rho$ and $0 < z \le \rho/128$, then $\delta_0 = \rho/16 \ge 8z$, so the largest J with $\delta_J \ge 8z$ exists and is finite. The error is at most $128z + 30000hD \le 30000\mathcal{Q}$.

4. If $\mathcal{Q} < \rho$ and $z = 0$, then $D > 0$ since the core has positive mass. The least J with $\delta_J \le hD$ exists and is finite; the error is less than $30001hD = 30001\mathcal{Q}$.

In the last two branches every local cap exists: the finite resolved law has positive gaps and $\eta_\chi(M) \to 0$. Moreover $\eta_\chi(4) = 1 - z > \delta_\ell/8$, including at the boundary $z = \rho/128$. Poisson means are positive and finite. Independently drawn counts and the uniform per-replication certificates give, for every q,

$$\mathbb{E}_q N \le \sum_{\ell=0}^{J} n_\ell k_\ell = B.$$

The counts are almost surely finite and all local caps are finite. Thus an actual admissible finite-level procedure exists at every $B > 0$; no attainment of a minimax infimum or infinite-level limit is assumed. It follows that $R(B) \le 30001\mathcal{Q}(B)$.

5.2 Lower-bound constants and separate hard parameters

Put $M_B = \min\{\rho, h\sqrt{\rho F}\}$ and

$$c_{\mathrm{ann}} = \frac{e^{-28}}{2000000}, \qquad k = \frac{\sqrt{c_{\mathrm{ann}}}}{128 \cdot 1024 \cdot 3584}.$$

In the packing branch, additive localization gives

$$R(B) \ge \frac{1}{128} \sum_i \min\{\theta_i/64, \sqrt{c_{\mathrm{ann}}}\, h\sqrt{\theta_i G_i}\} \ge kM_B.$$

The last step uses $\sqrt{c_{\mathrm{ann}}} < 1/64$, the proportional rarity allocation, and $\sum_i G_i \ge F/3584$. In the high-band branch $F \le 16C_H$ and $C_H = 4096(1 + \log 6) < 16384$. Hence the Bernoulli-label floor yields $R(B) \ge M_B/16384 \ge kM_B$ there as well.

If $h \ge t$, the tail bound gives $R \ge \mathcal{Q}/32$. If $h < t$ and $\lambda \ge t/4$, it gives $R \ge \mathcal{Q}/288$. Otherwise both $R \ge T_h/32$ and $R \ge kM_B$ hold. Since $F \le \Gamma_\kappa \le 4F$,

$$\mathcal{Q} \le T_h + 2M_B \le (32 + 2/k)R, \qquad R \ge \frac{k}{32k + 2}\mathcal{Q} \ge \frac{k}{3}\mathcal{Q}.$$

Here $32k \le 1$. The other branches have larger constants, so this proves one lower constant uniformly over all laws, rarities and budgets. The tail and core obstructions need not be witnessed by the same parameter: a worst-case risk exceeds each obstruction, hence their maximum. No exchange of a supremum and a sum is being used.

The explicit bound in Theorem 1.1 is therefore

$$10^{-20}\mathcal{Q}(B) \le R(B) \le 30001\mathcal{Q}(B).$$

Indeed $k/3 = e^{-14}/[\sqrt{2000000}\,(3 \cdot 128 \cdot 1024 \cdot 3584)]$, and $e < 3$, $\sqrt{2000000} < 1500$ and

$$3^{14}\, 1500\, (3 \cdot 128 \cdot 1024 \cdot 3584) < 10^{20}$$

certify this deliberately loose lower constant.

5.3 Lower-bound quantifiers and budget transport

The half-prior components are subsets of the full rarity class after the fractional mixture is integrated. Each induced endpoint and its population midpoint is admissible, so the original uniform budget applies to all three. Auxiliary oracle types strengthen only the lower experiment; they are not observations available to the upper estimator. The original algorithm is simulated using persistent private labels, and genuine oracle observations never exceed its original work.

At each fixed budget, the common-pair result selects a pair simultaneously for all local policies and deterministic caps. That pair is held fixed while the cap tends to infinity. Population midpoint work is bounded by B, and the overflow probabilities at the two endpoints sum to at most $2B/L$ at cap L. Neither a null-only nor an endpoint-only budget is substituted for the uniform assumption. The population midpoint is the experiment with an independent midpoint latent pair for every fresh coin, not the mixture of the two full population endpoint transcripts.

This completes the proof of Theorem 1.1.

6 Consequences

Write $c_0 = 10^{-20}$ and $C_0 = 30001$ for the constants in Theorem 1.1.

Corollary 6.1 (Mean-square form)**.** *For every law, rarity and budget in Theorem 1.1,*

$$c_0^2 \min\{\rho^2, T_h^2 + \rho\Gamma_\kappa/B\} \le R(B)^2 \le 2C_0^2 \min\{\rho^2, T_h^2 + \rho\Gamma_\kappa/B\}.$$

If $B \ge d_1^{-2}$ *and* $J_\nu = \int d^{-1}\nu(dd)$*, then*

$$c_0^2 \min\{\rho^2, (J_\nu^2 + \rho\Gamma_\kappa)/B\} \le R(B)^2 \le 2C_0^2 \min\{\rho^2, (J_\nu^2 + \rho\Gamma_\kappa)/B\}.$$

Proof. For $x, y \ge 0$, $x^2 + y^2 \le (x+y)^2 \le 2(x^2+y^2)$. Taking the minimum with ρ^2 preserves these comparisons, with the same factor two. Apply this to $x = T_h$ and $y = h\sqrt{\rho\Gamma_\kappa}$ and square (6). If $B \ge d_1^{-2}$, then $h \le d_i$ at every atom, so $T_h = hJ_\nu$ exactly, including at equality. □

Corollary 6.2 (One-gap specialization)**.** *For* $\mu = \delta_d$*,* $0 < d \le 1/2$*,*

$$R_{\delta_d,\rho}(B) \asymp \min\{\rho, \sqrt{\rho}/(d\sqrt{B})\}.$$

Proof. Here $\nu = \rho\delta_d$, $\kappa = (1-\rho)\delta_d$, and $\Gamma_\kappa = (1-\rho)/d^2$. Put $x = h/d$. If $x \ge 1$, then $\mathcal{Q}(B) = \rho$. If $x < 1$, the sum inside the minimum defining $\mathcal{Q}$ is $x[\rho + \sqrt{\rho(1-\rho)}]$. Since $0 < \rho \le 1/4$, the bracket lies between $(\sqrt{3}/2)\sqrt{\rho}$ and $(3/2)\sqrt{\rho}$. Taking the minimum with ρ and applying the theorem proves the comparison with universal constants. □

For a general finite law the core entropy vanishes only in the normalized entropy term when a single band is occupied; its base cost A^* is always positive. The quantile split can place part of one atom in both ν and κ. These cases are included in the theorem without any generic-position assumption.

The formulas concern a uniform expected observation budget. They do not supply a deterministic total cap, a high-confidence guarantee, or an estimator that learns an unknown magnitude law. The finite support and strictly positive gaps ensure that all caps used in the upper construction exist. Extending these assumptions requires additional arguments; Remark 7.4 records the extension to general laws that follows, conditionally, from [8].

7 Relation to the entropy–tail selection law

This section is deterministic: it compares the closed functional $\mathcal{Q}(B)$ with the selection functional of the companion general-law manuscript [8]. It is not used in the proof of Theorem 1.1.

7.1 The selection functional

Let μ be any probability law on $(0, 1/2]$, not necessarily finite. For such μ, define $R_{\mu,\rho}(B)$ by (2), with the model of Section 1 unchanged except that $D_k \sim \mu$, q ranges over measurable functions with $0 \le q \le 1$, and $\theta(q) = \int q\, d\mu$. Use the bands $I_j = (b_j/2, b_j]$, $b_j = 2^{-j-1}$, of (4) and the functional $\mathcal{G}$ of Section 2.3. For a finite measure $\sigma \le \mu$ put $a_j(\sigma) = \int_{I_j} d^{-2}\, \sigma(dd)$ and $A_\sigma = \sum_j a_j(\sigma)$, and define $\Gamma_\sigma = \sup_J \mathcal{G}(a_0(\sigma), \dots, a_J(\sigma))$, with values in $[0, \infty]$. When only finitely many $a_j(\sigma)$ are positive this is $\mathcal{G}(a(\sigma))$; for $\sigma = \kappa$ and a finite law it is the core entropy of (4). Since $\mathcal{G}$ is coordinatewise increasing, $\sigma \le \sigma'$ implies $\Gamma_\sigma \le \Gamma_{\sigma'}$, and $\Gamma_\sigma \ge A_\sigma$. Write $H_\sigma = \int d^{-1}\, \sigma(dd)$ and, with $h = B^{-1/2}$,

$$\mathcal{K}(B) = \min\left\{\rho,\ \inf_{0\le\sigma\le\mu}\left[h\sqrt{\rho\Gamma_\sigma} + \int \min\{1, h/d\}\,(\mu-\sigma)(dd)\right]\right\}. \tag{30}$$

This is the functional $\mathcal{K}_{\mu,\rho}(B)$ of [8]. That manuscript states and proves $R_{\mu,\rho}(B) \asymp \mathcal{K}_{\mu,\rho}(B)$ for every probability law on $(0, 1/2]$, every $0 < \rho \le 1/4$ and every $B > 0$, with explicit constants 10^{-17} and 33000. Its proof has not been externally refereed or formally verified, and it is not assumed anywhere in this manuscript.

For a general law, define the quantile split as in Section 1. Let $t = \min\{s : \mu((0, s]) \ge \rho\}$, which exists and is positive. Let ν be μ on $(0, t)$ plus the fraction of the atom at t, if any, that gives ν total mass ρ, and let $\kappa = \mu - \nu$. Then ν is supported on $(0, t]$, κ is supported on $[t, 1/2]$, and only finitely many bands meet $[t, 1/2]$. Define $T_h = \int \min\{1, h/d\}\, \nu(dd)$ and $\mathcal{Q}(B) = \min\{\rho, T_h + h\sqrt{\rho\Gamma_\kappa}\}$. For a finite law these are the quantities of (3) and (5).

7.2 Scale-separated cost

Besides (7), the comparison uses one more entropy bound.

Lemma 7.1 (Scale-separated cost)**.** *If $t > 0$ and ξ is a finite measure on $[t, 1/2]$, then $\Gamma_\xi \le 4H_\xi/t$.*

Proof. The bound is trivial for $\xi = 0$, so let $\xi \ne 0$. Let j_0 be the band containing t, so $t \le b_{j_0}$. A band meeting $[t, 1/2]$ has index $j_0 - r$ with $r \ge 0$ and consists of gaps in $(b_{j_0}2^{r-1}, b_{j_0}2^r]$. There every gap of ξ satisfies $d \ge t$ and $d > b_{j_0}2^{r-1} \ge t2^{r-1}$, so $d^{-1} \le 2^{1-r}/t$. Writing c_r and g_r for the inverse-square and inverse costs of ξ in that band, indexed by the offset r, we get $c_r \le 2^{1-r} g_r/t$, $\sum_r c_r = A_\xi$ and $\sum_r g_r = H_\xi$. Only finitely many bands are occupied, so $\Gamma_\xi = A_\xi(1+\mathrm{Ent}(c/A_\xi))$. Comparing the normalized costs with the probability vector $\pi_r = 2^{-(r+1)}$ gives $\mathrm{Ent}(c/A_\xi) \le \sum_r (c_r/A_\xi)\log(1/\pi_r)$, whence

$$\Gamma_\xi \le \sum_{r\ge0} c_r[1 + (r+1)\log 2] \le \frac{1}{t}\sum_{r\ge0} 2^{1-r}[1 + (r+1)\log 2]\, g_r.$$

The coefficient $2^{1-r}[1 + (r+1)\log 2]$ is largest at $r = 0$, where it equals $2 + 2\log 2 < 4$. □

7.3 The comparison

Proposition 7.2 (Quantile versus selection)**.** *For every probability law μ on $(0, 1/2]$, every $0 < \rho \le 1/4$ and every $B > 0$,*

$$\mathcal{K}(B) \le \mathcal{Q}(B) \le (1 + 2\sqrt{2})\, \mathcal{K}(B) < 4\, \mathcal{K}(B).$$

If $h \ge t$, then $\mathcal{K}(B) = \mathcal{Q}(B) = \rho$.

Proof. The measure $\sigma = \kappa$ is admissible in (30), and $\mu - \kappa = \nu$. Its objective is $h\sqrt{\rho\Gamma_\kappa} + T_h$, so $\mathcal{K}(B) \le \mathcal{Q}(B)$.

For the upper comparison fix $0 \le \sigma \le \mu$ with finite objective, let $f = d\sigma/d\mu \in [0,1]$, and write

$$x = h\sqrt{\rho\Gamma_\sigma}, \qquad y = \int \min\{1, h/d\}\,(\mu - \sigma)(dd), \qquad S = x + y.$$

Since $\nu \le \mu$, we have $(1-f)\nu \le \mu - \sigma$ and $f\nu \le \sigma$, so $\int f\,d\nu \le \rho$ and $\int d^{-2} f\,\nu(dd) \le A_\sigma \le \Gamma_\sigma$. With Cauchy–Schwarz,

$$T_h \le y + h\int d^{-1} f\,\nu(dd) \le y + h\sqrt{\rho \int d^{-2} f\,\nu(dd)} \le y + x = S.$$

As $T_h \le \rho$ always, $T_h \le \mathcal{K}(B)$. If $h \ge t$, every gap of ν has $\min\{1, h/d\} = 1$, so $T_h = \rho$ and therefore $\mathcal{K}(B) = \mathcal{Q}(B) = \rho$.

Now let $h < t$, and split $\kappa = f\kappa + (1-f)\kappa$. Both parts live on the finitely many bands meeting $[t, 1/2]$. By (7), monotonicity and Lemma 7.1,

$$\Gamma_\kappa \le 2\Gamma_{f\kappa} + 2\Gamma_{(1-f)\kappa} \le 2\Gamma_\sigma + \frac{8}{t} H_{(1-f)\kappa}.$$

Every gap of ν is at most t, so $\min\{1, h/d\} \ge h/t$ there and $h\rho/t \le T_h$. Every gap of κ is at least $t > h$, so $hH_{(1-f)\kappa} = \int (h/d)\,(1-f)\kappa(dd) \le y$. Hence

$$h\sqrt{\rho\Gamma_\kappa} \le \sqrt{2}\,x + \sqrt{8\,(h\rho/t)\,(hH_{(1-f)\kappa})} \le \sqrt{2}\,x + \sqrt{8T_h y},$$

and, with $T_h \le S$,

$$\mathcal{Q}(B) \le T_h + \sqrt{2}\,x + \sqrt{8T_h y} \le S + \sqrt{2}\,x + \sqrt{8Sy}.$$

If $S = 0$ then $\mathcal{Q}(B) = 0$. Otherwise put $u = y/S \in [0,1]$, so that $x = (1-u)S$ and

$$S + \sqrt{2}\,x + \sqrt{8Sy} = S\big[1 + \sqrt{2}\,(1 - u + 2\sqrt{u})\big] \le (1 + 2\sqrt{2})\,S,$$

because $1 - u + 2\sqrt{u} \le 2$ on $[0,1]$. Taking the infimum over σ, and using $\mathcal{Q}(B) \le \rho$, gives $\mathcal{Q}(B) \le (1+2\sqrt{2})\min\{\rho, \inf_\sigma S\} = (1+2\sqrt{2})\,\mathcal{K}(B)$. □

7.4 Consequences

Corollary 7.3 (Selection form for finite laws)**.** *For every finite law, rarity and budget in Theorem 1.1,*

$$10^{-20}\,\mathcal{K}(B) \le R_{\mu,\rho}(B) \le 30001(1+2\sqrt{2})\,\mathcal{K}(B) < 1.2 \cdot 10^5\,\mathcal{K}(B).$$

Proof. Combine Theorem 1.1 with Proposition 7.2. □

Corollary 7.3 is the finite-law case of the characterization stated in [8]; here it follows from Theorem 1.1, whose proof does not invoke the theorem of [8]. Conversely, that theorem, if granted, gives the rate $R \asymp \mathcal{Q}$ of Theorem 1.1 for finite laws through Proposition 7.2, without the separate argument of this paper. Its explicit constants 10^{-17} and 33000 then give $10^{-17}(1+2\sqrt{2})^{-1}\mathcal{Q} \le R \le 33000\,\mathcal{Q}$: a better lower constant than 10^{-20} and a slightly worse upper one than 30001.

The two results should be read as one rate established by two arguments, not as independent characterizations. Nor are the two arguments independent evidence: both manuscripts were developed from the same research notes. They share the observable upper construction (the upper bound here is the companion's construction applied to the fixed quantile split) and most appendix lemmas, and differ mainly in the lower-bound reduction.

Remark 7.4 (General laws, conditionally). Proposition 7.2 holds for every probability law on $(0, 1/2]$, and $\mathcal{Q}(B)$ is then still a closed expression: κ occupies finitely many bands. If $c\,\mathcal{K} \le R \le C\,\mathcal{K}$ holds for every such law, as [8] states with $c = 10^{-17}$ and $C = 33000$, then

$$\frac{c}{1+2\sqrt{2}}\,\mathcal{Q}(B) \le R_{\mu,\rho}(B) \le C\,\mathcal{Q}(B)$$

for every probability law on $(0, 1/2]$. This conditional extension is not a result of the present manuscript; it rests entirely on the theorem of [8].

The comparison was supplied by an external adversarial review (Astra Pro) of revision 1 of this manuscript, at repository commit `2984835`. Its proof used hard selections; the proof above works directly with the selection over submeasures of [8].

A Observable classification and finite Poisson allocation

This appendix proves the single-coin classification bounds and the finite multilevel estimator used in Section 4. Throughout this appendix, a false-positive probability means the joint event $\mathbb{P}_q(Z = 1, s = 0)$, with the binary sign $s = \mathbf{1}_{\{S=+1\}}$, not the conditional probability $\mathbb{P}_q(Z = 1 \mid s = 0)$. All likelihoods refer to the actual binary observations. A reference distribution evaluates that prefix and does not reveal or redraw the latent state of a coin.

A.1 A capped classifier that does not know the gap

For a dyadic integer $M \geq 1$, inspect prefix lengths $1, 2, 4, \ldots, M$. At a prefix $n < M$, stop if

$$\left|\widehat{p}_n - \frac{1}{2}\right| > t_{n,M} := \sqrt{\frac{1 + \log(M/n)}{n}},$$

and output the side of $1/2$ containing $\widehat{p}_n$. At M, output $\mathbf{1}_{\{\widehat{p}_M > 1/2\}}$, with a negative output on ties. Denote the output and work by $(Z_M^{\mathrm{ord}}, T_M^{\mathrm{ord}})$.

Lemma A.1 (Ordinary conditional error and work)**.** *For every $0 < d \leq 1/2$ and either latent sign,*

$$\mathbb{P}_{d,s}(Z_M^{\mathrm{ord}} \neq s) \leq \min\{1, (Md^2)^{-2}\}, \tag{31}$$

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} \leq o_M(d) := \min\{M, 17d^{-2}[1 + \log_+(Md^2)]\}. \tag{32}$$

Proof. For independent Bernoulli observations of mean p, Hoeffding's inequality gives $\mathbb{P}(\widehat{p}_n - p \geq t) \leq e^{-2nt^2}$ and the analogous lower-tail bound [5]. A wrong early output at a fixed inspected prefix requires a deviation of at least $d + t_{n,M}$. Applying a union bound on the underlying full tape, without conditioning on the stopping event, therefore gives

$$\mathbb{P}_{d,s}(Z_M^{\mathrm{ord}} \neq s) \leq e^{-2Md^2} + e^{-2} \sum_{\substack{n<M \\ n \text{ dyadic}}} (n/M)^2 e^{-2nd^2}.$$

For any dyadic sequence of positive u values, the intervals $[u/2, u]$ have disjoint interiors and

$$\int_{u/2}^{u} y e^{-2y}\, dy \geq \frac{3}{8} u^2 e^{-2u}.$$

Consequently $\sum_u u^2 e^{-2u} \leq 2/3$. Writing $x = Md^2 \geq 1$ and using $x^2 e^{-2x} \leq e^{-2}$, the preceding error bound is at most $(5/3)e^{-2}x^{-2} < x^{-2}$. For $x < 1$, the probability bound by one suffices.

Since the stopping time takes dyadic values,

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} = 1 + \sum_{\substack{n<M \\ n \text{ dyadic}}} n\, \mathbb{P}_{d,s}(T_M^{\mathrm{ord}} > n).$$

If $d > 2t_{n,M}$, survival implies a deviation from the true mean of at least $d/2$. Thus

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} \leq 1 + \sum_{\substack{n<M \\ n \text{ dyadic}}} n \left[\mathbf{1}_{\{d \leq 2t_{n,M}\}} + 2e^{-nd^2/2}\right].$$

For $x = Md^2 \geq 1$, the first indicator implies $nd^2 \leq 4(1 + \log x)$. Indeed, this is immediate if $nd^2 < 1$; otherwise $\log(M/n) = \log(x/(nd^2)) \leq \log x$. The corresponding dyadic sum is at most $8d^{-2}(1 + \log x)$. Also

$$\int_{u/2}^{u} e^{-y/2}\, dy \geq \frac{1}{2} u e^{-u/2} \quad \Longrightarrow \quad \sum_u u e^{-u/2} \leq 4.$$

The exponential part of the work bound is therefore at most $8d^{-2}$. Since $1 \le d^{-2}$, total expected work is at most $17d^{-2}(1+\log x)$. Finally $T_M^{\mathrm{ord}} \le M$ deterministically; when $x < 1$, the displayed minimum equals M. □

For any finite measure ξ on $(0, 1/2]$, write

$$\eta_\xi(M) = \int \min\{1, (Md^2)^{-2}\}\, \xi(dd), \qquad O_\xi(M) = \int o_M(d)\, \xi(dd).$$

Both are homogeneous in ξ. The first is nonincreasing in M and the second is nondecreasing. Pointwise comparison also gives $\eta_\xi(4M) \ge \eta_\xi(M)/16$.

Lemma A.2 (A homogeneous dyadic kernel estimate)**.** *For every finite measure ξ and integer $K \ge 1$,*

$$\sum_{k=1}^{K} \sqrt{\eta_\xi(4^k) O_\xi(4^k)} \le 208 \int \min\{2^K, d^{-1}\}\, \xi(dd).$$

In particular,

$$\sum_{k\ge 1} \sqrt{\eta_\xi(4^k) O_\xi(4^k)} \le 208 \int d^{-1}\, \xi(dd), \tag{33}$$

with extended-value interpretation.

Proof. Use bands $(2^{-j-1}, 2^{-j}]$, $j \ge 1$, with masses w_j, and put $c_j = 2^j w_j$. Direct substitution into the two kernels yields

$$\eta_\xi(4^k) \le 2^{-k} \left[\sum_{j\ge k} c_j 2^{k-j} + 16 \sum_{j<k} c_j 2^{3(j-k)} \right] = 2^{-k} A_k,$$

$$O_\xi(4^k) \le 2^k \left[\sum_{j\ge k} c_j 2^{k-j} + 136 \sum_{j<k} c_j 2^{j-k}(1+k-j) \right] = 2^k D_k.$$

For the second display on $j < k$, use $d^{-2} \le 4\, 2^{2j}$ and $1 + \log(4^k d^2) \le 2(1+k-j)$. On $j \ge k$, the deterministic cap gives the first sum exactly as an upper bound.

Apply $\sqrt{A_k D_k} \le (A_k + D_k)/2$ and sum over $k \le K$. For a fixed $j \le K$, the three relevant geometric sums obey

$$\sum_{k\le j} 2^{k-j} \le 2, \qquad \sum_{r\ge 1} 2^{-3r} = \frac{1}{7}, \qquad \sum_{r\ge 1} 2^{-r}(1+r) = 3.$$

The coefficient of c_j is thus at most $2 + 8/7 + 204 < 208$. For $j > K$, only the first kernels occur, with coefficient at most $2\, 2^{K-j}$. Hence the result is at most

$$208 \sum_{j\le K} 2^j w_j + 2\, 2^K \sum_{j>K} w_j.$$

For $j \le K$, $\min\{2^K, d^{-1}\} \ge 2^j$ on the band; for $j > K$ it equals 2^K. This proves the finite bound. Increasing K and using monotone convergence gives (33). All rearrangements involve nonnegative terms, so normalization and moment assumptions are unnecessary. □

A.2 A weighted bank compared with the full null mixture

For a binary prefix $x_{1:n}$, let $P_{p,n}(x_{1:n})$ be its probability under iid Ber(p) observations, and let $P_{\mathrm{f},n} = P_{1/2,n}$. Choose any nonempty finite or countable set of canonical bands

$$I_j = (b_j/2, b_j], \qquad b_j = 2^{-j-1},$$

and positive weights π_j summing to one on these bands. Set $r_j^- = b_j/2$ and $r_j^+ = b_j/4$, and define

$$L_{j,n}^- = \frac{P_{1/2-r_j^-,n}}{P_{\mathrm{f},n}}, \qquad M_n = \frac{\sum_j \pi_j P_{1/2+r_j^+,n}}{P_{-,\mu,n}}, \qquad P_{-,\mu,n} = \int P_{1/2-d,n}\,\mu(dd).$$

For $0 < \epsilon < 1$, stop negative when any $L_{j,n}^- \geq 1/(\epsilon\pi_j)$, and stop positive when $M_n \geq 1/(\rho\epsilon)$. Give negative crossings priority. A zero denominator triggers a positive stop. Denote the bank stopping time by T_{bank}; it may be infinite outside its selected bands.

Lemma A.3 (Full-mixture likelihood bank)**.** *The probability of a wrong bank stop is at most $2\rho\epsilon$ uniformly over $q \in \mathcal{Q}_\rho$, and its false-positive probability is at most $\rho\epsilon$. These statements concern the event that the bank stops and do not assume termination at every magnitude.*

For $d \in I_j$ in a selected band,

$$\mathbb{E}_{d,-} T_{\mathrm{bank}} \leq \frac{3}{4d^2}[1 + \log(1/\epsilon) + \log(1/\pi_j)]. \tag{34}$$

Suppose a finite number $V(d) > 0$ satisfies

$$\beta_\mu(n) := \int e^{-2nx^2}\,\mu(dx) \leq V(d) \qquad \text{for every integer } n \geq d^{-2}.$$

Then

$$\mathbb{E}_{d,+} T_{\mathrm{bank}} \leq \frac{6}{d^2}\left[1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+ \frac{V(d)}{\rho}\right]. \tag{35}$$

In particular the bank has finite conditional work on every selected band whenever such an envelope is supplied.

Proof. A nonnegative supermartingale Y_n with $Y_0 = 1$ has $\mathbb{P}(\sup_n Y_n \geq a) \leq 1/a$: stopping at the first crossing or a fixed horizon gives $1 \geq \mathbb{E}Y_{T\wedge m} \geq a\,\mathbb{P}(T \leq m)$, and increasing the horizon proves the claim. This elementary maximal inequality also works when the process is a supermartingale only on supported prefixes.

Under a positive coin of gap d, the next multiplier of $L_{j,n}^-$ has mean $1 - 4r_j^- d \leq 1$. Its negative-crossing probability is at most $\epsilon\pi_j$. A countable union bound gives conditional negative-crossing probability at most ϵ, so its positive-sign contribution under any admissible prior is at most $\rho\epsilon$.

Under the full all-negative mixture, M_n is a nonnegative supermartingale on supported prefixes. If a prefix x has positive null probability, conditional expectation sums only children xu with positive null probability; their numerator probabilities sum to at most the numerator at x. This proves the supermartingale inequality even when μ has mass at $d = 1/2$. A zero-denominator prefix has null probability zero. Thus a positive crossing has null-mixture probability at most $\rho\epsilon$. Its actual negative-sign contribution is no larger, since it multiplies the conditional crossing probabilities by $1 - q(d) \leq 1$. The two error claims follow.

For negative work, fix a selected-band gap $d \in I_j$ and $r = r_j^- \in [d/2, d]$, with $r \leq 1/4$. The matching log likelihood has mean increment

$$g(d, r) = \frac{1}{2}\log(1 - 4r^2) + d\log\frac{1 + 2r}{1 - 2r} \geq -\frac{8}{3}r^2 + 4dr \geq \frac{4}{3}d^2.$$

Here $\log(1-x) \geq -x/(1-x)$ and $\log((1+x)/(1-x)) \geq 2x$ establish the first lower bound; the concave quadratic takes the value $4d^2/3$ at both endpoints $r = d/2, d$. Every upward log increment is at most $\log(3/2) < 1$. For the first correct-reference crossing τ, the stopped-sum identity at $\tau \wedge m$ gives

$$g(d,r)\mathbb{E}(\tau \wedge m) = \mathbb{E}\log L^-_{j,\tau\wedge m} \leq 1 + \log(1/(\epsilon\pi_j)).$$

Monotone convergence proves integrability of τ and the stated work bound, since $T_{\text{bank}} \leq \tau$. No unbounded stopping identity is invoked before integrability is established.

For positive work, let $E_n = \{\widehat{p}_n - 1/2 \geq d/2\}$. On this event, for $x < 1/2$,

$$\log \frac{P_{1/2-x,n}}{P_{\text{f},n}} = \frac{n}{2}\log(1-4x^2) - n(\widehat{p}_n - \tfrac{1}{2})\log\frac{1+2x}{1-2x} \leq -2nx^2.$$

For $x = 1/2$ the likelihood is zero, since E_n entails at least one head. Therefore $P_{-,\mu,n}/P_{\text{f},n} \leq \beta_\mu(n)$. The matching positive reference $r = r_j^+ \in [d/4, d/2]$ satisfies, on the same event,

$$\log \frac{P_{1/2+r,n}}{P_{\text{f},n}} \geq n\left[-\frac{8}{3}r^2 + 2dr\right] \geq nd^2/3.$$

Set

$$U = 1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+(V(d)/\rho), \qquad n_0 = \left\lceil 3d^{-2}U \right\rceil.$$

For every $n \geq n_0$, E_n implies $M_n \geq \pi_j e^{nd^2/3}/V(d) \geq 1/(\rho\epsilon)$. Thus $T_{\text{bank}} > n$ implies E_n^c, whose positive-coin probability is at most $e^{-nd^2/2}$. Summing the survival probabilities yields

$$\mathbb{E}_{d,+}T_{\text{bank}} \leq 3d^{-2}U + 2 + 2d^{-2} \leq 6d^{-2}U.$$

The last inequality uses $U \geq 1$ and $d^2 \leq 1/4$.

For a countable bank, the positive mixture is a consistent probability law and the negative stopping event is a measurable countable union. At any fixed time only finitely many negative references can cross: $L^-_{j,n} \leq (3/2)^n$ requires $\pi_j \geq 1/[\epsilon(3/2)^n]$ for a crossing. All preceding arguments therefore apply to finite and countable banks alike. □

A.3 Racing procedures and retaining a genuine unit

Lemma A.4 (Observable race)**.** *Suppose two classifiers run on the odd and even subsequences of one unit's observation tape, with independent auxiliary seeds. Stop the physical procedure when either classifier stops and use its output. On completed counterfactual subsequences its work satisfies*

$$T_{\text{race}} \leq 2\min\{T_1, T_2\}.$$

Its error event is contained in the union of the component wrong-stop events. If either component has a finite deterministic cap, the race terminates with a finite deterministic cap.

Proof. By physical time $2n$, each component has received n observations. If either component would stop after n of its own observations, the race stops no later than $2n$. This proves the pathwise work bound, including an infinite stopping time for the other component. The selected output is wrong only if the component that produces it makes a wrong stop. Conditional on the latent (D, S), both subsequences are iid Bernoulli tapes of the correct mean, so all component error and work bounds apply. The policy uses only the physical observed prefix and its seeds. □

A.4 Finite multilevel allocation at every positive budget

Lemma A.5 (Independent Poisson allocation)**.** *Let $Z_0, \dots, Z_J \in \{0,1\}$ be observable single-unit classifiers, each with finite deterministic cap. Suppose uniformly over $q \in \mathcal{Q}_\rho$ that their error probabilities are at most e_ℓ, their false-positive probabilities are at most f_ℓ, and their expected work is at most $C_\ell > 0$. Put*

$$v_0 = \rho + f_0, \qquad v_\ell = e_\ell + e_{\ell-1} \quad (\ell \geq 1), \qquad k_0 = C_0, \qquad k_\ell = C_\ell + C_{\ell-1} \quad (\ell \geq 1).$$

Assume $v_\ell > 0$ and set $A_J = \sum_{\ell=0}^{J} \sqrt{v_\ell k_\ell}$. For every $B > 0$ there is an observable population estimator of uniform expected work at most B and worst RMS error at most

$$\sqrt{e_J^2 + A_J^2/B}. \tag{36}$$

Proof. A level-zero correction uses a fresh genuine unit and returns $\Delta_0 = Z_0$. For $\ell \geq 1$, a correction draws one fresh latent unit, runs Z_ℓ on its observations, then runs $Z_{\ell-1}$ with a fresh seed on the next unused observations of the same unit, and returns $\Delta_\ell = Z_\ell - Z_{\ell-1}$.

Conditional on the latent state, the unused suffix after a finite stopping time is an iid tape with the same Bernoulli mean: for each possible stopped prefix, independence of its subsequent coordinates gives the claim, and summing over the stopped prefixes preserves it. The second classifier therefore has its original marginal distribution under the same latent state. No new magnitude or sign is drawn between the two classifiers. Their disagreement implies that at least one misclassifies that common sign, and hence

$$\mathbb{E}_q \Delta_\ell^2 \leq v_\ell, \qquad \mathbb{E}_q W_\ell \leq k_\ell, \qquad \sum_{\ell=0}^{J} \mathbb{E}_q \Delta_\ell = \mathbb{E}_q Z_J.$$

At level zero use $\mathbb{P}_q(Z_0 = 1) \leq \mathbb{P}_q(s = 1) + f_0 \leq \rho + f_0$.

Before drawing the genuine units and their observation tapes, draw the Poisson replication counts independently across levels, with means

$$n_\ell = \frac{B}{A_J} \sqrt{\frac{v_\ell}{k_\ell}}.$$

For each replication use an independent genuine unit, and return

$$\widehat{\theta} = \sum_{\ell=0}^{J} \frac{1}{n_\ell} \sum_{i=1}^{N_\ell} \Delta_{\ell,i}, \qquad N_\ell \sim \text{Poisson}(n_\ell).$$

Its mean is $\mathbb{E}_q Z_J$, so its absolute bias is at most e_J. The expected work is at most $\sum_\ell n_\ell k_\ell = B$.

For independent marks Δ and a Poisson count of mean n, conditioning on the count gives $\text{Var}(\sum_{i=1}^{N} \Delta_i) = n\mathbb{E}\Delta^2$. Consequently the independent level variances sum to at most

$$\sum_{\ell=0}^{J} \frac{v_\ell}{n_\ell} = \frac{A_J^2}{B}.$$

The divisor is the deterministic mean n_ℓ, including when the realized count is zero; it is not the random count N_ℓ. Count fluctuations contribute the squared mark mean, which is why the second moment rather than just the mark variance appears. The mark and its observation cost may be dependent within a replication; the count is independent of the entire sequence of these paired draws.

For any positive allocation x_ℓ with $\sum_\ell k_\ell x_\ell \leq B$, Cauchy–Schwarz gives

$$\left(\sum_\ell \sqrt{v_\ell k_\ell} \right)^2 \leq \left(\sum_\ell v_\ell / x_\ell \right) \left(\sum_\ell k_\ell x_\ell \right).$$

Thus the displayed allocation minimizes this second-moment certificate under its expected-work constraint. This is not a deterministic work cap. Bias squared plus variance proves (36). All counts are almost surely finite, all component caps are finite, and all first and subsequent observations are included in W_ℓ. □

The allocation may use Poisson means below one; this is valid for every $B > 0$. The bound is uniform over the original rarity class. Finite-level bounds can be optimized by taking an infimum over finite constructions without asserting the existence of a single infinite-level algorithm.

B One-unit null-work obstructions

This appendix proves the sign-testing and annular ingredients used in Section 3. The weighted sign-testing mechanism is an adaptation of Chen–Li–Qiao's gap-entropy argument, including their Bernoulli extension [2, Lemma 4.1 and Remark D.4]. We give the full finite-gap version needed here, followed by the scale-sensitive gate and its actual minimum-band-mass constraint.

B.1 Weighted sign testing

Lemma B.1 (Weighted sign entropy)**.** *Let $d_1 > \cdots > d_K > 0$, with $d_j/d_{j+1} \geq 2$ and $d_1 \leq 1/128$. One observable binary sign test, with the same independent seed law at all gaps, has error at most $1/10$ under every Bernoulli law $1/2 \pm d_j$. Suppose its stopping time τ has finite expectation under both signs at every such gap. Put $c_j = d_j^2 \mathbb{E}_{-d_j}\tau$. For every probability vector π on these gaps,*

$$\sum_{j=1}^{K} \pi_j c_j \geq \frac{1 + \mathrm{Ent}(\pi)}{257}. \tag{37}$$

The stopping and decision rules are the same measurable functions of the seed and observation tape at every gap; the realized gap is not supplied as an input. The test may depend on the entire known finite gap list and on the weights. Zero weights are allowed, with $0\log(1/0) = 0$. No termination or correctness assumption under the fair coin is required.

Proof. First, the ordinary stopped change-of-measure inequality gives a uniform positive floor for each c_j. The terminal sign decision has minus/plus KL divergence at least

$$\mathrm{kl}(9/10, 1/10) = \frac{4}{5}\log 9 > \frac{8}{5}.$$

One Bernoulli observation has divergence at most

$$\frac{(2d_j)^2}{(1/2 + d_j)(1/2 - d_j)} \leq \frac{64}{3} d_j^2.$$

The stopped chain rule, justified by finite expected endpoint work, implies $c_j > 3/40 > 1/16$. More explicitly, apply the chain rule first to the transcript capped at N. Encode a nonterminal decision by a third symbol. Its probability tends to zero at both endpoints, so lower semicontinuity of finite-alphabet KL and monotone convergence of $\mathbb{E}_{-d_j}(\tau \wedge N)$ justify the stopped inequality. This step uses no fair-law stopping assumption.

Set $b_0 = 1/1024$ and

$$n_j = \left\lfloor b_0/d_j^2 \right\rfloor, \qquad u_j = \left\lceil 4\mathbb{E}_{-d_j}\tau \right\rceil.$$

Since $d_j \leq 1/128$, we have $n_j \geq b_0/(2d_j^2) \geq 8$. The KL divergence between the complete prefixes of length n_j at the minus and plus laws is at most $1/48$. These prefixes include the independent

seed and unused bits if the test has already stopped. Pinsker's inequality therefore gives total variation less than $1/8$. An early minus is a plus-law error, so its probability under the minus law is at most $1/10 + 1/8$. Markov's inequality gives $\mathbb{P}_{-d_j}(\tau > u_j) \le 1/4$. Consequently the event

$$E_j = \{\text{output minus},\ n_j < \tau \le u_j\}$$

has $\mathbb{P}_{-d_j}(E_j) > 1/4$.

Let $\mathbb{P}_0$ denote the fair-coin law with the same seed and policy. Comparing deterministic prefixes of length u_j gives KL at most $4d_j^2 u_j \le 16c_j + 1$. Binary data processing and $\mathrm{kl}(p, z) \ge p\log(1/z) - \log 2$ yield

$$\mathbb{P}_0(E_j) \ge \exp(-64c_j - 8). \tag{38}$$

Only deterministic finite prefixes have been compared with the fair law.

We next bound a weighted overlap pathwise. On a path with finite terminal time t, participation in E_j implies

$$t > n_j \ge \frac{b_0}{2d_j^2}, \qquad t \le 4\mathbb{E}_{-d_j}\tau + 1.$$

In particular $t \ge 2$ and $c_j \ge (t-1)d_j^2/4 \ge td_j^2/8$. Order the participating gaps from smallest to largest. Their squared values grow by at least a factor of four, and the smallest has $td_j^2 > b_0/2$. The corresponding costs are thus bounded below by $b_0 4^r/16 = 4^{r-7}$, for $r = 0, 1, \ldots$. Hence

$$\sum_j \frac{\mathbf{1}_{E_j}}{1 + c_j} \le \sum_{r \ge 0} \frac{1}{1 + 4^{r-7}} \le 7 + \sum_{r \ge 7} 4^{7-r} = \frac{25}{3}.$$

On a nonterminating fair path no E_j occurs, so the same inequality holds. Integrating and using (38), together with $1 + c_j \le e^{c_j}$, gives

$$Z := \sum_j e^{-65c_j} \le (25/3)e^8, \qquad \log Z < 11.$$

Gibbs' inequality for π and the normalized masses e^{-65c_j}/Z implies

$$\mathrm{Ent}(\pi) \le 65 \sum_j \pi_j c_j + \log Z.$$

Writing $\bar{c} = \sum_j \pi_j c_j$ and using $\bar{c} \ge 1/16$, we obtain $1 + \mathrm{Ent}(\pi) \le 65\bar{c} + 12 \le 257\bar{c}$, as required. □

B.2 A moment comparison on the half-prior class

Write $X_\rho = \{0 \le q \le 1/2 : \int q\, d\mu \le \rho\}$. For an integrable observable one-unit output Y with finite uniform variance and uniform bias at most $\rho/64$, put $V = \sup_{q \in X_\rho} \mathrm{Var}_q\, Y$ and $M = \sup_{q \in X_\rho} \mathbb{E}_q Y^2$.

Lemma B.2 (Moment comparison)**.** *For $0 < \rho \le 1/4$, one has $V \ge 2\rho/3$ and $M \le 3V/2$.*

Proof. Let $f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y$. Comparing the null population with the constant prior $q = \rho$ gives $\int f\, d\mu \ge 31/32$. Conditional variance at this constant prior and Jensen's inequality yield

$$V \ge \rho(1-\rho) \int f^2\, d\mu \ge \rho(1-\rho)(31/32)^2 \ge 2\rho/3.$$

Every mean has absolute value at most $65\rho/64$, so $M \le V + (65\rho/64)^2 \le 3V/2$. The constant prior also gives integrability of both fixed-sign second moments; all conditional statements may be taken outside a common null set. □

B.3 An activation gate that respects the magnitude scale

The gate argument charges activation at the information scale $a^2\mathbb{E}_0 T$. In this subsection μ is any probability law supported on $(0, a]$, $a \le 1/4$, and

$$\mathcal{X}_\rho = \{q : 0 \le q \le \tfrac{1}{2},\ \textstyle\int q\, d\mu \le \rho\}.$$

Let (Y, T) be an observable one-coin procedure with uniform bias at most $\rho/64$, finite uniform variance V, and finite expected work under every $q \in \mathcal{X}_\rho$. Write

$$M_0 = \sup_{q \in \mathcal{X}_\rho} \mathbb{E}_q Y^2, \qquad W_0 = \mathbb{E}_0 T, \qquad P = M_0 W_0.$$

The moment comparison in Lemma B.2 gives $M_0 \le 3V/2$.

More explicitly, observability means that an independent seed U and the genuine observations generate $\mathcal{F}_n = \sigma(U, X_1, \dots, X_n)$, that T is a nonnegative integer-valued stopping time, and that Y is $\mathcal{F}_T$-measurable. The procedure has no access to the latent magnitude or sign. Restarts below refresh U and the unused observation block, while retaining that same latent state. These are the observation-model assumptions of Section 1.

Lemma B.3 (A scale-sensitive baseline and gated activation)**.** *Put*

$$C_g = e^7, \qquad \kappa = 64/3, \qquad c_g = (256\kappa C_g^4)^{-1}.$$

Every procedure above satisfies

$$P \ge \frac{\rho}{64a^2}. \tag{39}$$

If $a^2 P \le c_g$, there is an observable procedure Z on the same one-coin experiment, with moment bound m_Z, null work W_0^Z, and a measurable set $\mathcal{G}$ of good magnitudes, such that

$$\begin{gathered} \sup_{q \in \mathcal{X}_\rho} \mathbb{E}_q Z^2 \le m_Z, \qquad \rho/16 \le m_Z \le 2C_g^4 a^2 P, \\ m_Z W_0^Z \le C_g^4 P, \qquad \mu(\mathcal{G}^c) \le 144 m_Z. \end{gathered} \tag{40}$$

At every good gap, both signs have finite conditional expected work and

$$|\mathbb{E}[Z \mid d, 0]| \le \frac{1}{8}, \qquad \mathbb{E}[Z \mid d, 1] \ge \frac{1}{2}, \qquad \mathbb{E}[Z^2 \mid d, 0] \le \frac{m_Z}{\rho}, \qquad \mathbb{E}[Z^2 \mid d, 1] \le \frac{3m_Z}{\rho}. \tag{41}$$

One common sign test with both errors below $1/10$ on these gaps uses

$$n = \lceil 1024 m_Z/\rho \rceil \le 1040 m_Z/\rho \tag{42}$$

independent restarts of Z on unused observations of the same coin. Its negative expected work is $n\,\mathbb{E}[T_Z \mid d, 0]$.

Proof. We first prove (39). Let P_- and P_+ be the full transcript laws with uniformly negative and uniformly positive signs, respectively. The priors $q = 0$ and $q \equiv \rho$ give

$$\mathbb{E}_+ Y - \mathbb{E}_- Y \ge 31/32, \qquad \mathbb{E}_- Y^2 \le M_0, \qquad \mathbb{E}_+ Y^2 \le M_0/\rho.$$

The positive moment and expected work are finite because $q \equiv \rho$ is admissible. Revealing D for the information calculation, and using the stopped chain rule, gives

$$\mathfrak{h}(P_-, P_+) \le \tfrac{1}{2}\kappa a^2 W_0,$$

where $\mathfrak{h} = 1 - \mathrm{Aff}$. Indeed the one-observation endpoint divergence is at most κd^2 for $d \le 1/4$. For any square-integrable output F,

$$|\mathbb{E}_P F - \mathbb{E}_Q F|^2 \le 4(\mathbb{E}_P F^2 + \mathbb{E}_Q F^2)\mathfrak{h}(P, Q). \tag{43}$$

Thus

$$(31/32)^2 \le (5\kappa/2)a^2 P/\rho,$$

which implies (39).

For the gate, on a full observation tape define

$$S_n = \sum_{i=1}^{n}(2X_i - 1), \qquad n_0 = \lceil a^{-2} \rceil, \qquad H_0 = \lceil a^{-1} \rceil,$$

$$G = \min\{n_0, \inf\{n \ge 1 : |S_n| \ge H_0\}\}, \qquad \sigma = T \wedge G, \qquad E = \{T \ge G\}. \tag{44}$$

The event E is measurable at σ, including the tie $T = G$. Include the entire independent seed in the stopped prefix. At σ, $|S_\sigma| \le H_0$ and $\sigma \le n_0$. For a fixed $d \le a$ and sign s, the likelihood relative to the fair stopped-prefix law is

$$\exp\left\{(2s-1)S_\sigma \operatorname{arctanh}(2d) + \frac{\sigma}{2}\log(1 - 4d^2)\right\}.$$

The inequalities

$$\operatorname{arctanh}(2d) \le 8a/3, \qquad -\tfrac{1}{2}\log(1 - 4d^2) \le 8a^2/3$$

show that the absolute logarithm is at most $37/6 < 7$. Every fixed-state prefix law, and the all-negative magnitude mixture, therefore has density in $[C_g^{-1}, C_g]$ relative to fair.

Let $p = \mathbb{P}_{\mathrm{fair}}(E)$ and $p_s(d) = \mathbb{P}_{d,s}(E)$. The number p is determined by the known policy, its seed law, and a bounded fair-prefix experiment. It does not depend on the unknown prior or genuine latent state. This is an existence argument; no efficient computation of arbitrary policy probabilities is claimed. No termination under the fair law beyond this bounded prefix is needed. Then

$$p/C_g \le p_s(d) \le C_g p. \tag{45}$$

Bounded fair stopping gives $\mathbb{E}_{\mathrm{fair}} S_\sigma^2 = \mathbb{E}_{\mathrm{fair}}\sigma$. A gate activation either reaches n_0 or hits H_0, so

$$p \le \left(n_0^{-1} + H_0^{-2}\right)\mathbb{E}_{\mathrm{fair}}\sigma \le 2a^2\mathbb{E}_{\mathrm{fair}}\sigma \le 2C_g a^2 W_0. \tag{46}$$

Set $Y_e = Y\mathbf{1}_{\{T<G\}}$, explicitly zero on E. This is a function of the bounded prefix. Its second moment under either fixed state is at most $C_g^2 M_0$. The stopped endpoint divergence and the same prefix domination give

$$\mathrm{KL}(P_{-d}^\sigma, P_{+d}^\sigma) \le \kappa d^2 \mathbb{E}_{-d}\sigma \le \kappa C_g^2 a^2 W_0.$$

Since $\mathfrak{h} \le \mathrm{KL}/2$, (43) implies

$$|\mathbb{E}_{+d} Y_e - \mathbb{E}_{-d} Y_e|^2 \le 4\kappa C_g^4 a^2 P. \tag{47}$$

Under $a^2 P \le c_g$, every such early mean difference is at most $1/8$ in absolute value. In this branch $p > 0$: if $p = 0$, the gate never activates under any state, so $Y = Y_e$, contradicting the integrated mean separation $31/32$ from $q = 0, q \equiv \rho$.

Here is an actual observable procedure realizing the conditioning. Keep the same genuine coin. Repeatedly run the original procedure, using a fresh original seed and unused iid flips for each

trial. Reject a trial if its original stopping time is before its gate. When a trial reaches the gate, accept it, continue that original trial to its original stopping time, and return

$$Z = C_g pY.$$

Every rejected observation is charged. Conditional on d, s, trials are iid with acceptance probability $p_s(d) > 0$. Writing $w_s(d) = \mathbb{E}[T \mid d, s]$, the renewal identity gives

$$\mathbb{E}[T_Z \mid d, s] = \frac{w_s(d)}{p_s(d)}. \tag{48}$$

This identity includes the work of all rejected trials and the final accepted trial. To verify it without assuming independence of acceptance and trial cost, let L be the first accepted trial. Conditional on (d, s),

$$\mathbb{E}\sum_{k=1}^{L} T_k = \sum_{k\geq 1} \mathbb{E}[T_k \mathbf{1}_{\{L\geq k\}}] = \sum_{k\geq 1}(1 - p_s(d))^{k-1} w_s(d) = w_s(d)/p_s(d).$$

The event $\{L \geq k\}$ involves only preceding trials, so it is independent of the current trial cost. It is not necessary that T_k be independent of its own acceptance event. Moreover,

$$\mathbb{E}[Z^2 \mid d, s] = \frac{C_g^2 p^2}{p_s(d)} \mathbb{E}[Y^2 \mathbf{1}_E \mid d, s] \leq C_g^3 p\, \mathbb{E}[Y^2 \mid d, s].$$

With $m_Z = C_g^3 p M_0$, integration and (46)–(48) give

$$\sup_q \mathbb{E}_q Z^2 \leq m_Z, \qquad m_Z \leq 2C_g^4 a^2 P, \qquad W_0^Z \leq C_g W_0/p, \qquad m_Z W_0^Z \leq C_g^4 P. \tag{49}$$

Expected work under every q is finite by the same comparison.

We establish conditional sign separation directly; conditioning has not preserved the estimator's means. Write

$$f_s(d) = \mathbb{E}[Y \mid d, s], \quad \ell_s(d) = \mathbb{E}[Y \mathbf{1}_E \mid d, s], \quad z_s(d) = \mathbb{E}[Z \mid d, s].$$

The set $\{f_+ - f_- < 3/4\}$ has mass below 2ρ. Otherwise, writing that set as S with $\mu(S) \geq 2\rho$, the prior $q = (\rho/\mu(S))\mathbf{1}_S$ lies in $\mathcal{X}_\rho$ and has total mass ρ. It would contradict the bias guarantee and the mean separation $31\rho/32$. This uses fractional prior values, so it applies unchanged when μ has atoms. Outside that set, (47) gives $\ell_+ - \ell_- \geq 5/8$. Also

$$z_s(d) = \frac{C_g p}{p_s(d)} \ell_s(d), \qquad 1 \leq \frac{C_g p}{p_s(d)} \leq C_g^2.$$

If $|z_-(d)| \leq 1/8$, then $|\ell_-(d)| \leq 1/8$, $\ell_+(d) \geq 1/2$, and $z_+(d) \geq 1/2$. The exceptional mass for $|z_-| > 1/8$ is at most $64 m_Z$. Since

$$m_Z \leq 2C_g^4 c_g = 1/(128\kappa) < 1/256,$$

this preliminary mean-good set has mass at least $1 - 2\rho - 64 m_Z \geq 1/4$. Taking $q \equiv \rho$ in the second-moment bound yields $m_Z \geq \rho/16$.

Finally, put $a_Z(d) = \mathbb{E}[Z^2 \mid d, 0]$ and $b_Z(d) = \mathbb{E}[Z^2 \mid d, 1]$. The set $\{a_Z > m_Z/\rho\}$ has mass at most ρ. The set $\{b_Z - a_Z > 2m_Z/\rho\}$ has mass below 2ρ, again by using a fractional admissible prior on a set of mass at least 2ρ. Removing these sets and null sets of infinite conditional work gives total exceptional mass

$$5\rho + 64 m_Z \leq 144 m_Z$$

and proves (41). Average n independent restarts of Z on unused blocks of the same coin, and threshold at $5/16$. Both mean margins are at least $3/16$, so Chebyshev gives errors at most

$$\frac{3m_Z/\rho}{n(3/16)^2} \leq 1/12 < 1/10.$$

The bound on n follows from $m_Z \geq \rho/16$, and work adds, proving (42). □

B.4 A scaled annular obstruction

For a positive vector $v = (v_i)$, set

$$\mathcal{G}(v) = A[1 + \mathrm{Ent}(v/A)], \qquad A = \sum_i v_i.$$

A useful elementary identity is

$$\frac{\partial \mathcal{G}}{\partial v_i} = 1 + \log(A/v_i) \geq 1. \tag{50}$$

Thus this cost-entropy functional is coordinatewise increasing. In particular, if $u \leq v_i \leq U$ for K coordinates, then

$$uK(1 + \log K) \leq \mathcal{G}(v) \leq UK(1 + \log K). \tag{51}$$

Proposition B.4 (A gated annular lower bound)**.** *Suppose μ is supported on $(0, a]$, $a \leq 1/4$. Take a nonempty finite annular family*

$$I_j = (b_j/2, b_j], \qquad b_j/b_{j+1} \geq 4, \qquad b_1 \leq 1/128,$$

with positive masses $w_j = \mu(I_j)$. Define

$$w_* = \min_j w_j, \qquad A = \sum_j w_j/b_j^2, \qquad \pi_j = \frac{w_j/b_j^2}{A}, \qquad \Gamma^\circ = A[1 + \mathrm{Ent}(\pi)].$$

Every one-coin procedure with the assumptions above satisfies

$$V\mathbb{E}_0 T \geq \frac{e^{-28}}{2\,000\,000} \min\{\rho\Gamma^\circ, w_*/a^2\}. \tag{52}$$

Proof. If $a^2 P \geq c_g$, use $w_* \leq 1$ and $M_0 \leq 3V/2$. Otherwise apply Lemma B.3. If $m_Z > w_*/288$, then

$$P > w_*/(576 C_g^4 a^2).$$

If $m_Z \leq w_*/288$, at least half the mass of every annulus is good. Averaging selects $d_j \in I_j \cap \mathcal{G}$ with

$$w_-^Z(d_j) \leq \frac{4}{w_j} \int_{I_j} w_-^Z(d)\, \mu(dd).$$

The selected gaps are separated by a factor greater than two and bounded by $1/128$. The representative points may depend on the procedure being bounded. This is legitimate: the weighted sign lemma holds for every such finite list. After the points have been selected, the same already-constructed restart test is run at all of them; it is not retuned to the selected point. The weights remain the normalized band costs π_j, not the original band probabilities. Apply Lemma B.1 to the common restart test, with weights π_j:

$$\frac{1 + \mathrm{Ent}(\pi)}{257} \leq n \sum_j \pi_j d_j^2 w_-^Z(d_j) \leq \frac{4n}{A} W_0^Z.$$

Consequently,

$$\rho\Gamma^\circ \leq 1028 \rho n W_0^Z \leq 1\,069\,120\, m_Z W_0^Z \leq 1\,069\,120\, C_g^4 P.$$

Converting $P = M_0 W_0$ to $V W_0$ completes all branches with the displayed conservative constant. No atomlessness is used. □

The entropy branch is available when $\rho a^2 \Gamma^\circ \leq w_*$. This condition is invariant under a common rescaling of all magnitudes.

C Convex comparison, common pairs, and additive localization

This appendix proves the specialized transfer inputs used by the scalar lower bound. The finite-experiment convex-estimation principle has antecedents in Juditsky–Nemirovski and Polyanskiy–Wu [3, 4]; the midpoint adaptive decomposition is due to Lee–Valiant [1]. The arguments needed for this model are included in full.

C.1 Finite categorical convex comparison

Lemma C.1 (Finite categorical convex comparison)**.** *Let X be a nonempty compact convex subset of a finite-dimensional real vector space. Let $\ell : X \to \mathbb{R}$ be affine, and let $p(x)$ be a strictly positive categorical probability vector with affine continuous coordinates. For $n \geq 1$ independent observations from $p(x)$, define*

$$\Omega_n = \max\{\ell(x) - \ell(y) : x, y \in X, \mathrm{Aff}(p(x), p(y))^n \geq 1/2\}.$$

For every $\eta > 0$, there is a bounded additive estimator $F = c + \sum_{i=1}^n \phi_i(Z_i)$ satisfying

$$\sup_{x \in X} \{\mathbb{E}_x (F - \ell(x))^2\}^{1/2} \leq \tfrac{3}{2}\Omega_n + \eta. \tag{53}$$

Proof. We reproduce the short argument needed for the numerical constant. Set $s = \log 2$, let $\mathcal{F}$ be the finite-dimensional space of additive functions, and define, for $\alpha > 0$,

$$\Phi(x, y; \phi, \alpha) = \ell(x) - \ell(y) + \alpha \log \mathbb{E}_y e^{\phi/\alpha} + \alpha \log \mathbb{E}_x e^{-\phi/\alpha} + 2\alpha s.$$

The function is continuous and concave in (x, y): independence writes each log moment as a sum of logarithms of positive affine forms. It is convex in (ϕ, α) by the perspective transformation of a log moment generating function. Sion's minimax theorem [6, corollary following Theorem 3.4, p. 174] therefore applies to $X \times X$ and $\mathcal{F} \times (0, \infty)$. Only $X \times X$ needs to be compact; the open, unbounded α domain is convex and causes no problem in Sion's one-compact-set form. The infimum in (ϕ, α) need not be attained, and the later argument uses a strict slack $R > S$. Write $2S = \inf_{\phi, \alpha > 0} \max_{x,y} \Phi$.

For a fixed pair, the half log likelihood ratio belongs to $\mathcal{F}$ and Cauchy–Schwarz gives

$$\inf_{\psi \in \mathcal{F}} [\log \mathbb{E}_y e^{\psi} + \log \mathbb{E}_x e^{-\psi}] = 2 \log \mathrm{Aff}(p(x)^{\otimes n}, p(y)^{\otimes n}).$$

Substituting $\phi = \alpha\psi$ and minimizing over $\alpha > 0$ gives the value $\ell(x) - \ell(y)$ when the affinity is at least $1/2$, and $-\infty$ otherwise. Consequently $2S = \Omega_n$. The constrained maximum exists because its feasible set is nonempty and compact.

For $R > S$, choose (ϕ, α) with $\max \Phi \leq 2R$. On the diagonal, Cauchy–Schwarz implies $\Phi(x, x; \phi, \alpha) \geq 2\alpha s$, hence $\alpha \leq R/s$. Set

$$U = \max_x \{\ell(x) + \alpha \log \mathbb{E}_x e^{-\phi/\alpha}\}, \qquad V = \max_x \{-\ell(x) + \alpha \log \mathbb{E}_x e^{\phi/\alpha}\}.$$

Then $U + V + 2\alpha s \leq 2R$. The estimator $F = \phi + (U - V)/2$ satisfies

$$\mathbb{E}_x e^{\pm(F - \ell(x))/\alpha} \leq e^{R/\alpha - s}.$$

Therefore $\mathbb{P}_x\{|F - \ell(x)| > R + t\} \leq e^{-t/\alpha}$ for $t \geq 0$. Tail integration yields

$$\mathbb{E}_x (F - \ell(x))^2 \leq R^2 + 2R\alpha + 2\alpha^2 \leq R^2 \left(1 + \frac{2}{\log 2} + \frac{2}{(\log 2)^2}\right) < 9R^2.$$

Let $R \downarrow S$, with the slack chosen for η. Every component is bounded because its alphabet is finite. This proves (53). □

C.2 Midpoint decomposition for interleaved observations

We use $\mathfrak{h}(P,Q) = 1 - \mathrm{Aff}(P,Q)$, so $0 \le \mathfrak{h} \le 1$ and $\mathrm{TV}(P,Q) \le \sqrt{2\mathfrak{h}(P,Q)}$. The following finite-prefix argument is the midpoint decomposition of Lee–Valiant [1, Lemma 17], written in the form needed here.

Lemma C.2 (Midpoint work decomposition)**.** *Fix two endpoint priors q_-, q_+ and their midpoint $\bar{q} = (q_- + q_+)/2$. Suppose every capped randomized one-unit policy A obeys*

$$\mathfrak{h}(P_-^A, P_+^A) \le \kappa\, \mathbb{E}_{\bar{q}} T_A.$$

Then every population algorithm with a deterministic total cap satisfies

$$\mathfrak{h}(P_-^{\mathrm{pop}}, P_+^{\mathrm{pop}}) \le \kappa\, \mathbb{E}_{\bar{q}} T_{\mathrm{pop}}.$$

It suffices that the local hypothesis hold up to that same total cap. If instead $q_\pm \le 1/2$ and the local bound uses null work $\lambda \mathbb{E}_0 T_A$, the conclusion holds with $\kappa = 2\lambda$. The statement also holds for finitely many observable component types, each with its own endpoint pair and midpoint, when the same coefficient bounds every local policy for every type. Types may be selected adaptively, but each fresh unit's type is fixed before its first observation and retained on every revisit.

Proof. Fix total cap K and condition on the algorithm's independent seed. Defer latent draws until a unit is first observed, which changes no observable law. Index units by that first observation. A consistent global leaf z contains local binary words $h_i(z)$, with empty words for unused slots. The endpoint leaf densities factor as

$$p_\pm^{\mathrm{pop}}(z) = \prod_{i=1}^{K} p_\pm(h_i(z)).$$

The common policy-consistency indicator is implicit. Put $m_i = (p_-(h_i) + p_+(h_i))/2 = p_{\bar{q}}(h_i)$ and $c_i = \sqrt{p_-(h_i) p_+(h_i)}$. Then

$$\prod_i m_i - \prod_i c_i \le \sum_i (m_i - c_i) \prod_{j \ne i} m_j.$$

The integral of $\prod_i m_i$ over consistent leaves is one, since it is the population law in which every fresh unit has midpoint prior. Integrating the inequality therefore bounds population squared Hellinger by a sum of hybrid squared Hellinger distances. In hybrid i only unit i has an endpoint prior; all other units have midpoint priors.

Simulate all other units from independent midpoint tapes included in an auxiliary seed, and run the population rule with one genuine unit in slot i. This is a capped local policy. Its full transcript, including that seed, determines the hybrid global transcript. Data processing bounds hybrid distance by local distance. The expected midpoint local works sum to $\mathbb{E}_{\bar{q}} T_{\mathrm{pop}}$ because each purchased observation belongs to exactly one slot. This proves the first assertion, including after integrating the original seed.

If $q_\pm \le 1/2$, the midpoint latent law dominates half the null latent law. The same measure inequality holds after passing through any observation policy and auxiliary seed. Thus $\mathbb{E}_0 T_A \le 2\mathbb{E}_{\bar{q}} T_A$.

For observable types, the type selected for a slot is a function of the history before its first observation. Use the resulting type-specific kernel in each leaf product. To implement hybrid slot i, first generate all observations before that slot under the componentwise midpoint law. Conditional on this prehistory and its selected type, the genuine unit has that type's prescribed endpoint law and is fresh, independently of the remaining auxiliary randomness. This is conditional independence: the latent law may depend on the type contained in the seed.

Retain the already generated tapes of other units on every revisit. For later slots, an independent array of potential tapes indexed by slot and type can be placed in the auxiliary seed; choose the appropriate unused tape only after that slot's type has been selected. The seed's distribution is the same under the two endpoint laws for the genuine unit. Conditional on the prehistory, the local hypothesis for its selected type applies. If slot i is never reached, the genuine local work is zero. Summing and integrating counts every purchased observation once under the actual componentwise midpoint population. No latent magnitude or sign of the genuine unit is revealed to the local policy. □

For unbounded stopping in the applications here, the same endpoint pair works at every cap, and the original uniform budget gives expected work at most B at both endpoints and at their admissible midpoint. Let N be original purchased work, and cap after L observations, recording an overflow flag if needed. Any terminal event restricted to $\{N \le L\}$ is measurable from this capped transcript. Therefore

$$\mathrm{TV}(P_-, P_+) \le \mathrm{TV}(P_-^{(L)}, P_+^{(L)}) + \mathbb{P}_-(N > L) + \mathbb{P}_+(N > L) \le \sqrt{2\kappa B} + 2B/L.$$

Letting $L \to \infty$ gives the terminal testing bound. For an oracle simulation, cap at the original observation count; its genuine work is no larger, and the same argument applies. This avoids exchanging a cap limit with a separately chosen pair.

C.3 The half-prior class and null work

Put

$$X_\rho = \left\{ q : 0 \le q \le \tfrac{1}{2}, \quad \theta(q) := \int q \, d\mu \le \rho \right\}, \qquad 0 < \rho \le \tfrac{1}{4}.$$

For a capped observable one-coin procedure with bounded real output Y, write

$$V = \sup_{q \in X_\rho} \mathrm{Var}_q \, Y, \qquad M = \sup_{q \in X_\rho} \mathbb{E}_q Y^2, \qquad W_0 = \mathbb{E}_0 T,$$

and define

$$\mathfrak{J}_0(b) = \inf \left\{ VW_0 : \sup_{q \in X_\rho} |\mathbb{E}_q Y - \theta(q)| \le b, \ T \text{ capped}, \ Y \text{ bounded} \right\}. \tag{54}$$

An empty infimum is infinity. The function is nonincreasing in b, and $\mathfrak{J}_0(\rho/2) = 0$.

C.4 Transfer at an arbitrary bias tolerance

Theorem C.3 (Bias-indexed common pairs and population lower bounds)**.** *Suppose that every capped bounded one-coin output with uniform bias at most $b > 0$ over X_ρ satisfies $VW_0 \ge J$, where $0 < J < \infty$. Then, for every $0 < \delta \le b/3$, there is one pair $q_-, q_+ \in X_\rho$ with target separation at least δ such that simultaneously for all capped randomized local policies A,*

$$\mathfrak{h}(P_{q_-}^A, P_{q_+}^A) \le 42 \frac{\delta^2}{J} \mathbb{E}_0 T_A. \tag{55}$$

Consequently,

$$R_{\mu,\rho}(B) \ge \frac{1}{128} \min\{b, \sqrt{J/B}\}. \tag{56}$$

The conclusion includes fully adaptive creation, revisiting, interleaving, nonlinear outputs and unbounded stopping.

Proof. The premise forces $b < \rho/2$, since the zero-work constant $\rho/2$ would otherwise contradict it. First fix a finite strictly positive categorical statistic of a capped policy of null work $w > 0$. For n independent runs, let $F = c + \sum_{i=1}^{n} Y_i$ be a bounded additive estimator of worst RMS r. Put $b_i = \mathbb{E}_0 Y_i$, $b_F = \mathbb{E}_0 F$, and choose an independent uniform index I. Running just its component and returning $Z = n(Y_I - b_I) + b_F$ preserves all population means at null work w.

The laws under q and zero share a subprobability of mass $1 - \theta(q)$, obtained by weighting the negative-sign kernel by $1 - q(d)$. The elementary common-mixture variance inequality gives

$$(\mathbb{E}_q Y_i - b_i)^2 \le \frac{2\theta(q)}{1-\theta(q)}(\operatorname{Var}_q Y_i + \operatorname{Var}_0 Y_i).$$

Indeed, either mean's displacement from the normalized common part is at most the square root of $\theta/(1-\theta)$ times its standard deviation; square the sum of the two displacements. Independence in F now yields

$$\sup_q \operatorname{Var}_q Z \le nr^2 \left(1 + \frac{4\rho}{1-\rho}\right) \le \tfrac{7}{3} nr^2.$$

If $r < b$, apply the assumed obstruction to Z; if $r \ge b$, retain that inequality. In both cases,

$$r \ge \min\{b, \sqrt{3J/(7nw)}\}.$$

Lemma C.1 gives an additive estimator of RMS at most $(3/2)\Omega_n + \eta$, where Ω_n is the maximum target separation whose product affinity is at least $1/2$. Letting $\eta \downarrow 0$ therefore gives

$$\Omega_n \ge \tfrac{2}{3} \min\{b, \sqrt{3J/(7nw)}\}.$$

Set $x = J/(21\delta^2 w)$. If $x \ge 2$, take $n = \lfloor x \rfloor$; then $\Omega_n \ge 2\delta$, and the corresponding pair satisfies

$$\mathfrak{h} \le \frac{\log 2}{n} \le 42 \log 2 \, \frac{\delta^2 w}{J} \le 42 \frac{\delta^2 w}{J}.$$

If $x < 2$, the last right side exceeds one, so any separated pair suffices. If $w = 0$, the seed-only first-observation event has probability zero; all local laws agree and $q_- = 0, q_+ = \delta$ suffices.

We now obtain one pair simultaneously for every capped local policy. For an arbitrary capped policy, partition its full transcript into finitely many measurable cells, then replace the cell independently by a uniform cell with probability $0 < \varepsilon < 1$. Run the policy even on replacement outcomes, so its null work is unchanged. The resulting strictly positive category probabilities and the target have compact convex finite-dimensional range, so the preceding comparison applies. For any finite collection of these experiments, independently mix them and reveal the chosen index. Categories with zero mixture weight can be dropped, leaving a strictly positive finite alphabet. For a fixed one-unit policy, its categorical law is affine in q: condition on the one genuine latent state first. This affineness is not being asserted for an entire adaptive population transcript.

Set $k = 42\delta^2/J$ and use the oriented compact convex pair set

$$K_\delta = \{(q_-, q_+) \in X_\rho^2 : \theta(q_+) - \theta(q_-) \ge \delta\}.$$

For experiment i, let w_i be its null work and $f_i(q_-, q_+) = \mathfrak{h}(P_{q_-}^{A_i}, P_{q_+}^{A_i}) - k w_i$. Each f_i is continuous and convex on K_δ. For any mixture weights λ, the revealed index gives exactly $\mathfrak{h}_{\mathrm{mix}} = \sum_i \lambda_i \mathfrak{h}_i$ and $w_{\mathrm{mix}} = \sum_i \lambda_i w_i$. The preceding one-experiment argument therefore implies

$$\min_{z \in K_\delta} \sum_i \lambda_i f_i(z) \le 0 \qquad \text{for every } \lambda \text{ in the finite simplex.}$$

Applying convex minimax in its indicated direction gives

$$\min_{z\in K_\delta}\max_i f_i(z)=\max_{\lambda}\min_{z\in K_\delta}\sum_i \lambda_i f_i(z)\le 0.$$

The left minimum is attained. Thus the finite family has a single feasible pair; separate pairs for the individual experiments would not suffice. All its constraint sets are closed in K_δ. The finite intersection property supplies one pair for all experiments. The intersection ranges over all caps, finite partitions and positive noise levels at once, so this pair does not depend on the stopping cap. The pair is fixed while $\varepsilon\downarrow 0$; finite categorical Hellinger distance is continuous even when some limiting cells have zero mass. Full squared Hellinger distance is the supremum over finite measurable partitions. For completeness, put $R=P+Q$ and $t=dP/dR$. Partition the range of t into intervals of width at most $1/m$. Since $g(t)=\sqrt{t(1-t)}$ satisfies $|g(u)-g(v)|\le\sqrt{|u-v|}$, the full and coarsened affinities differ by at most $2/\sqrt{m}$. Thus the supremum statement holds without a countable-generation assumption on the transcript space. This proves (55) for arbitrary transcript spaces. The quantifier is one pair for every policy and cap at fixed (μ,ρ,b,J,δ). The pair may change when δ, and hence the chosen budget-dependent separation, changes.

The midpoint decomposition in Lemma C.2 bounds population squared Hellinger by $84\delta^2B/J$: the local null cost is at most twice the midpoint cost, and summing midpoint costs counts each observation once. Set

$$\delta=\frac{1}{32}\min\{b,\sqrt{J/B}\}.$$

The distance is at most $84/1024<1/8$, hence TV is at most $1/2$. The same pair works for every deterministic observation cap. Finite expected endpoint work and increasing capped prefixes give the same terminal TV bound for unbounded stopping. Thresholding an estimator at the target midpoint gives RMS at least $\delta/4$, proving (56). A budget over the full rarity class implies the required budget over X_ρ. □

C.5 A direct sum of conditional experiments

Let $\sigma_1,\dots,\sigma_m$ be nonzero finite submeasures with $\sum_i\sigma_i\le\mu$. Write $\tau_i=\sigma_i((0,1/2])$ and $\nu_i=\sigma_i/\tau_i$. Disjoint restrictions are a special case; fractional restrictions of atoms are also allowed. Choose

$$0<r_i\le 1/4,\qquad \sum_i\tau_i r_i\le\rho.$$

Within component i, use the half-prior subclass

$$X_{r_i}(\nu_i)=\{0\le q\le 1/2:\ \textstyle\int q\,d\nu_i\le r_i\}.$$

The original population lower bound is valid on this subclass of the full rarity class.

Theorem C.4 (Additive localization)**.** *Suppose that, for each i, every capped bounded observable one-unit output with uniform bias at most $b_i>0$ over $X_{r_i}(\nu_i)$ satisfies*

$$\left(\sup_q \mathrm{Var}_q\,Y\right)\mathbb{E}_{i,0}T\ge J_i>0.$$

Then, for every $B>0$,

$$\boxed{R_{\mu,\rho}(B)\ge\frac{1}{128}\sum_{i=1}^m\tau_i\min\{b_i,\sqrt{J_i/B}\}.}\tag{57}$$

The same total budget B occurs in every summand.

Proof. Theorem C.3 applies within every component. Set

$$\delta_i = \frac{1}{32}\min\{b_i, \sqrt{J_i/B}\}.$$

It supplies a pair $q_{i,-}, q_{i,+} \in X_{r_i}(\nu_i)$, oriented to have conditional target separation at least δ_i, such that simultaneously for every capped local policy A,

$$\mathfrak{h}(P^A_{i,-}, P^A_{i,+}) \le 42\frac{\delta_i^2}{J_i}\mathbb{E}_{i,0}T_A \le \frac{42}{1024B}\mathbb{E}_{i,0}T_A. \tag{58}$$

Fix an arbitrary original algorithm with uniform expected budget B. Consider an oracle experiment that can select a component at no cost and then receive a genuine coin with magnitude law ν_i. Its component label is visible, while its magnitude and sign remain hidden. The algorithm can adaptively select, revisit, and interleave these coins. Apply the argument to the simulation of the fixed original algorithm described below. At the componentwise midpoint it induces the original parameter $\bar{q} = (q_- + q_+)/2 \in \mathcal{Q}_\rho$, so its genuine midpoint work is at most the original work and hence at most B. Under the conditional midpoint prior, $q \le 1/2$, so the midpoint law dominates half the null law. Lemma C.2 therefore bounds terminal squared Hellinger by $84/1024 < 1/8$: the local coefficient in (58) is common to all components, and the summed conditional midpoint costs count each purchased observation once. Increasing capped prefixes handles unbounded stopping. The total target separation is at least $\sum_i \tau_i\delta_i$.

Every original algorithm can be simulated in this stronger experiment. Privately choose the component label with probabilities τ_i whenever it creates a fresh coin. On the remaining probability mass, simulate the known all-negative residual law for free. Keep the chosen label when the coin is revisited. For fractional submeasures, integrating this private mixture label gives the original admissible parameters

$$q_\pm(d) = \sum_i \frac{d\sigma_i}{d\mu}(d)q_{i,\pm}(d), \qquad 0 \le q_\pm \le 1/2, \qquad \int q_\pm\, d\mu \le \sum_i \tau_i r_i \le \rho.$$

The number of genuine oracle observations does not exceed the original number. Thus the oracle lower bound applies to the original experiment. The Hellinger bound gives TV at most $1/2$, and midpoint testing gives RMS at least one quarter of the target separation. This is (57). □

For a direct explicit input, suppose component i is supported in $(0, a_i]$, $a_i \le 1/4$, and select a finite separated family of small-gap annuli. Let Γ_i° and $w_{*,i}$ denote its conditional cost entropy and minimum band mass. Proposition B.4 gives

$$b_i = r_i/64, \qquad J_i = c_{\mathrm{ann}}\min\{r_i\Gamma_i^\circ, w_{*,i}/a_i^2\}, \qquad c_{\mathrm{ann}} = \frac{e^{-28}}{2{,}000{,}000}.$$

Consequently, writing the corresponding unnormalized quantities with respect to σ_i as $\Gamma^\circ_{\sigma_i} = \tau_i\Gamma_i^\circ$ and $w_{*,\sigma_i} = \tau_i w_{*,i}$, we obtain

$$R_{\mu,\rho}(B) \gtrsim \sum_i \min\left\{\tau_i r_i, \sqrt{\frac{\min\{\tau_i r_i\Gamma^\circ_{\sigma_i}, \tau_i w_{*,\sigma_i}/a_i^2\}}{B}}\right\}. \tag{59}$$

For clarity, the squared component mass is essential before cancellation:

$$\tau_i^2 J_i = c_{\mathrm{ann}}\min\{\tau_i r_i\Gamma^\circ_{\sigma_i}, \tau_i w_{*,\sigma_i}/a_i^2\}.$$

This explains the extra τ_i in the saturation term of (59); it must not be dropped when passing between conditional and unnormalized quantities. Section 3 applies this bound to the components constructed by the quantile-core filter and packing.

References

[1] J. C. H. Lee and P. Valiant. *Uncertainty about Uncertainty: Optimal Adaptive Algorithms for Estimating Mixtures of Unknown Coins*. Proceedings of the 2021 ACM–SIAM Symposium on Discrete Algorithms (SODA), 414–433, 2021. https://arxiv.org/abs/1904.09228v3.

[2] L. Chen, J. Li, and M. Qiao. *Towards Instance Optimal Bounds for Best Arm Identification*. Proceedings of the 2017 Conference on Learning Theory, Proceedings of Machine Learning Research 65:535–592, 2017. https://proceedings.mlr.press/v65/chen17b.html.

[3] A. B. Juditsky and A. S. Nemirovski. *Nonparametric estimation by convex programming*. Annals of Statistics 37(5A):2278–2300, 2009. https://doi.org/10.1214/08-AOS654.

[4] Y. Polyanskiy and Y. Wu. *Dualizing Le Cam's method for functional estimation, with applications to estimating the unseens*. arXiv:1902.05616v2, 2021. https://arxiv.org/abs/1902.05616v2.

[5] W. Hoeffding. *Probability Inequalities for Sums of Bounded Random Variables*. Journal of the American Statistical Association 58(301):13–30, 1963. https://doi.org/10.1080/01621459.1963.10500830.

[6] M. Sion. *On general minimax theorems*. Pacific Journal of Mathematics 8(1):171–176, 1958. https://doi.org/10.2140/pjm.1958.8.171.

[7] P. M. Aronow, N. Kallus, and P. Lopatto. *A positive resolution of the gap-entropy conjecture*. arXiv:2609.10529v1, 2026. https://arxiv.org/abs/2609.10529v1.

[8] *A general entropy–tail law for adaptive rare-event estimation*. Companion manuscript, revised 29 October 2026. https://github.com/davidiach/adaptive-threshold-estimation/blob/29848354c638ccc2618ff8f9335544a0f1ff89f6/paper/rare-event/v2/main.pdf.